

■ BRAKE CONTROL SYSTEM

(Multi-terrain ABS with EBD, Brake Assist, A-TRC, VSC, Hill-start Assist Control, Downhill Assist Control and CRAWL)

1. General

The brake control system (Multi-terrain ABS with EBD, brake assist, A-TRC, VSC, hill-start assist control, downhill assist control, CRAWL) of new Land Cruiser (Station Wagon) has the following functions:

Function	Outline
Multi-terrain ABS (Anti-lock Brake System)	The ABS helps prevent the wheels from locking when the brakes are applied firmly or when braking on a slippery surface. In addition to the existing ABS function, the Multi-terrain ABS automatically calculates the optimum ABS control in accordance with the off-road situation such as a dirt or sandy road.
EBD (Electronic Brake force Distribution)	The EBD utilizes Multi-terrain ABS, realizing the proper brake force distribution between front and rear wheels in accordance with the driving conditions. In addition, during cornering braking, it also controls the brake forces of right and left wheels, helping to maintain the vehicle behavior.
Brake Assist	The primary purpose of the brake assist is to provide an auxiliary brake force to assist the driver who cannot generate a large brake force during emergency braking, thus helping draw the vehicle's brake performance.
A-TRC (Active Traction Control)	During rugged off-road driving, this function controls the engine output and the brake fluid pressure that is applied to the slipping wheel, and distributes the drive force that would have been lost through the slippage to the remaining wheels in order to achieve a LSD (Limited Slip Differential) effect. As a result, the vehicle's off-road drivability and ability to free itself from the mogul have been increased.
VSC (Vehicle Skid Control)	The VSC system helps prevent the vehicle from slipping sideways as a result of strong front wheel skid or strong rear wheel skid during cornering.
Hill-start Assist Control	When the vehicle starts off on a steep or a slippery hill, hill-start assist control detects the backward descent of the vehicle and effects 4-wheel hydraulic pressure control to reduce the backward speed of the vehicle. As a result, the vehicle can be started off on a hill more easily.
Downhill Assist Control*1	When the DAC switch is pressed with transfer in low range and without accelerator and brake pedals operation, downhill assist control activates to effect 4-wheel hydraulic pressure control, in order to maintain a constant low vehicle speed without causing the wheels to become locked. Thus, the vehicle can descend a steep hill in a stable manner.
CRAWL*2	When the vehicle runs on extremely bumpy rough ground or a slippery road, by activating the CRAWL, the engine output and the hydraulic pressure of the 4 wheel are regulated by the CRAWL, allowing the driver to drive the vehicle at the desired speed and keep that speed with virtually no accelerator or brake pedal operation. This allows the driver to concentrate on steering and maintains vehicle stability.

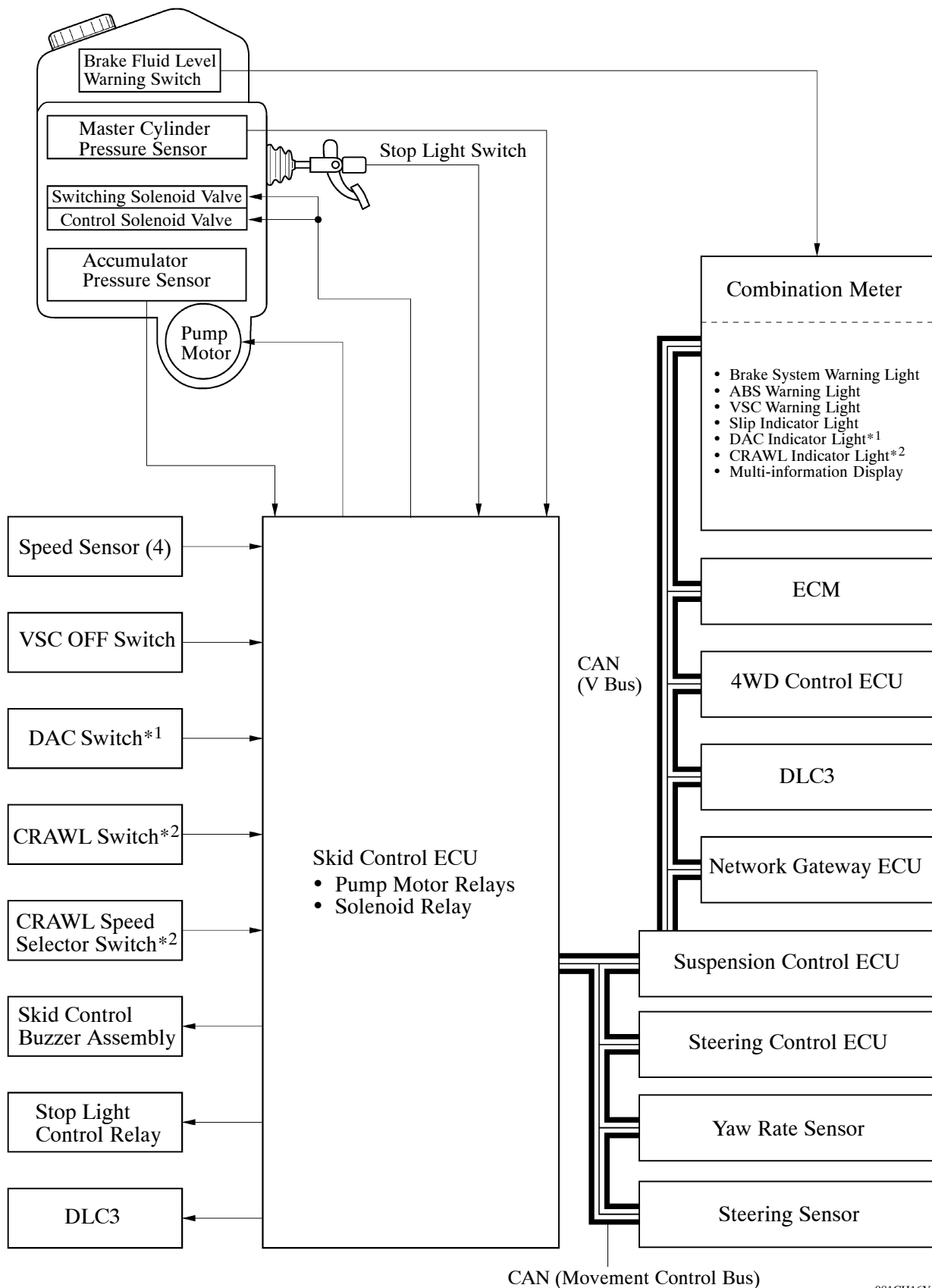
*1: Only for 1VD-FTV engine models

*2: Only for 2UZ-FE engine models

Service Tip

When the brake control system is activated, the brake pedal could shudder, which is a normal occurrence of the system in operation and should not be considered a malfunction.

► System Diagram ◀



*1: Only for 1VD-FTV engine models

*2: Only for 2UZ-FE engine models

2. Outline of Multi-terrain ABS

The detailed outline is the same as that of brake control system (Multi-terrain ABS). For details, see page CH-164.

3. Outline of EBD

The detailed outline is the same as that of brake control system (EBD). For details, see page CH-165.

4. Outline of Brake Assist

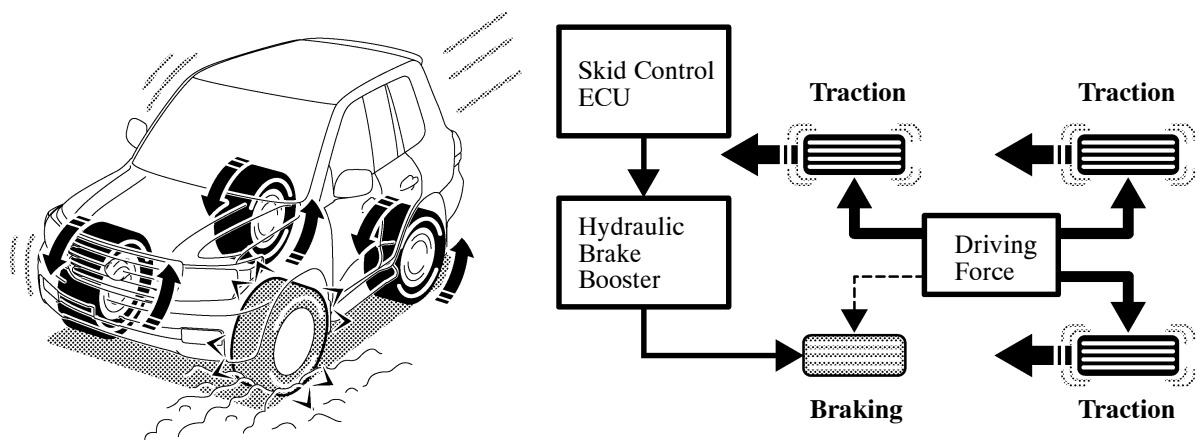
The detailed outline is the same as that of brake control system (brake assist). For details, see page CH-166.

5. Outline of A-TRC

General

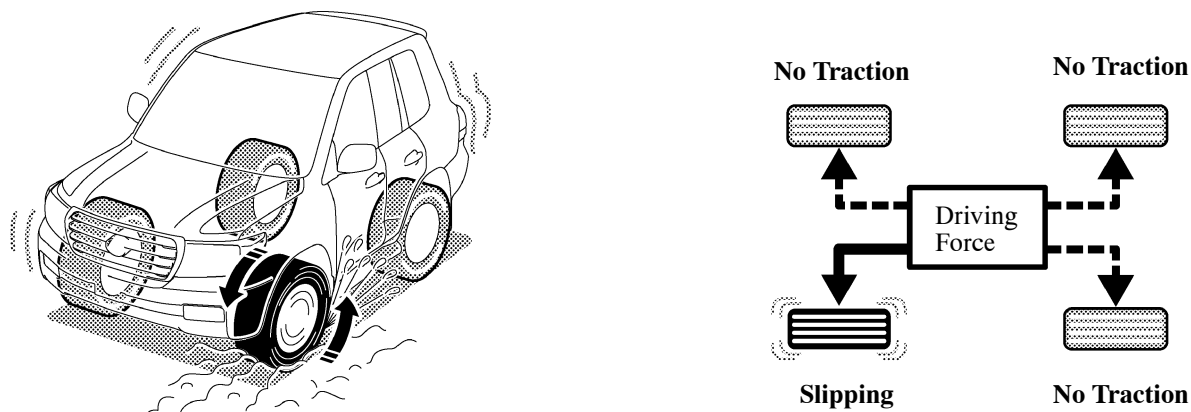
- If a tire slips while the vehicle is being driven on a snow-covered road or off-road, the function of the differential gear causes a large amount of drive force to be applied to the tire that is slipping.
- The A-TRC function helps restrain the slippage by controlling the engine output and brake fluid pressure that is applied to the slipping wheel, and distributes the drive force that would have been lost through the slippage to the remaining wheels in order to achieve an effect that is similar to the LSD (Limited Slip Differential).
- It independently controls the brake hydraulic pressure to the 4-wheel in accordance with the extent of the slippage at the wheels, as detected by the skid control ECU.

► with A-TRAC ◀



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► without A-TRAC ◀



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Effective of A-TRC

The effectiveness of the A-TRC is as follows:

- Off-road drivability that is equivalent to having the center differential locked and limited slip differential on both front and rear differentials has been realized.
- This function made the operation of the differential lock switches basically unnecessary to ensure the ease of driving.
- While realizing the off-road drivability that is equivalent to having the center and rear differentials locked, as compared to the differential gear in the locked state, the essential function of the differential gear itself ensures the ease of nimble steerability.

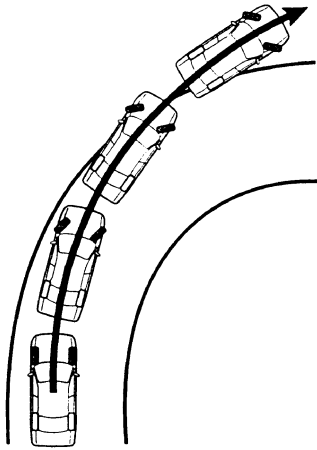
6. Outline of VSC

General

The followings are two examples that can be considered as circumstances in which the tires exceed their lateral grip limit.

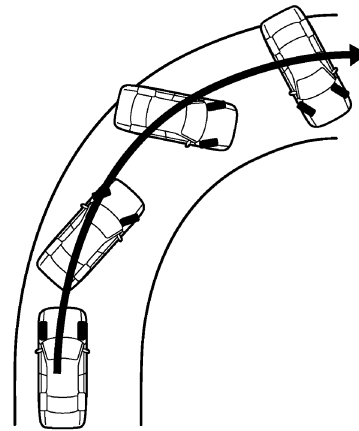
The VSC system is designed to help control the vehicle behavior by controlling the engine output and the brakes at each wheel when the vehicle is under one of the conditions indicated below.

- When the front wheels lose grip in relation to the rear wheels (front wheel skid tendency).
- When the rear wheels lose grip in relation to the front wheels (rear wheel skid tendency).



151CH17

Front Wheel Skid Tendency



189CH100

Rear Wheel Skid Tendency

Method for Determining the Vehicle Condition

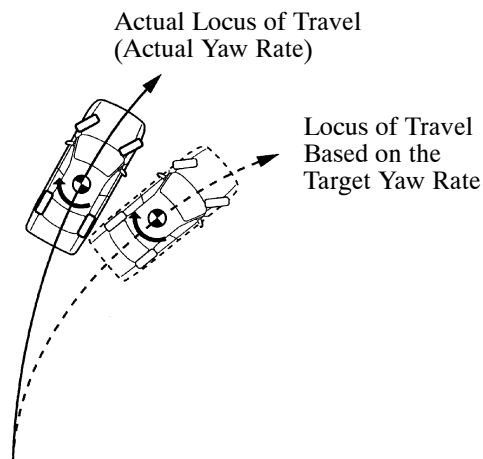
To determine the condition of the vehicle, sensors detect the steering angle, vehicle speed, vehicle's yaw rate, and the vehicle's lateral acceleration, which are then calculated by the skid control ECU.

1) Determining Front Wheel Skid

Whether or not the vehicle is in the state of front wheel skid is determined by the difference between the target yaw rate and the vehicle's actual yaw rate.

When the vehicle's actual yaw rate is smaller than the yaw rate (a target yaw rate that is determined by the vehicle speed and steering angle) that should be rightfully generated when the driver operates the steering wheel, it means the vehicle is making a turn at a greater angle than the locus of travel.

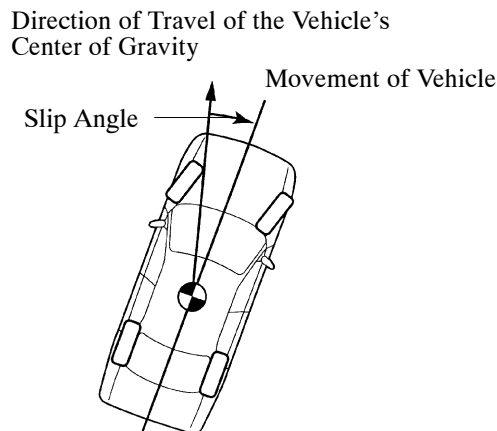
Thus, the skid control ECU determines that there is a large tendency to front wheel skid.



151CH19

2) Determining Rear Wheel Skid

Whether or not the vehicle is in the state of rear wheel skid is determined by the values of the vehicle's slip angle and the vehicle's slip angular velocity (time-dependent changes in the vehicle's slip angle). When the vehicle's slip angle is large, and the slip angular velocity is also large, the skid control ECU determines that the vehicle has a large rear wheel skid tendency.



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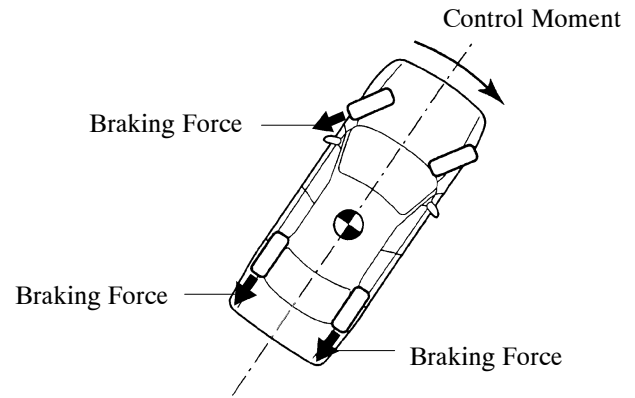
Method for VSC Operation

When the skid control ECU determines that the vehicle exhibits a tendency to front wheel skid or rear wheel skid, it decreases the engine output and applies the brake of a front or rear wheel to control the vehicle's yaw moment.

The basic operation of the VSC is described below. However, the control method differs depending on the vehicle's characteristics and driving conditions.

1) Dampening a Front Wheel Skid

When the skid control ECU determines that there is a large front wheel skid tendency, it counteracts in accordance with the extent of that tendency. The skid control ECU controls the engine power output and applies the brakes of the front wheel of the outer circle in the turns and rear wheels in order to restrain the front wheel skid tendency.



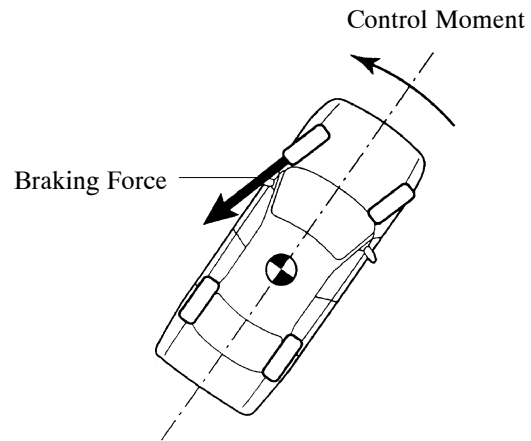
Making a Right Turn

189CH101

2) Dampening a Rear Wheel Skid

When the skid control ECU determines that there is a large rear wheel skid tendency, it counteracts in accordance with the extent of that tendency. It applies the brakes of the front wheel of the outer circle of the turn, and generates an outward moment of inertia in the vehicle, in order to restrain the rear wheel skid tendency. Along with the reduction in the vehicle speed caused by the braking force, the vehicle's stability is further improved.

In some cases, the skid control ECU applies the brake of the rear wheels, as necessary.

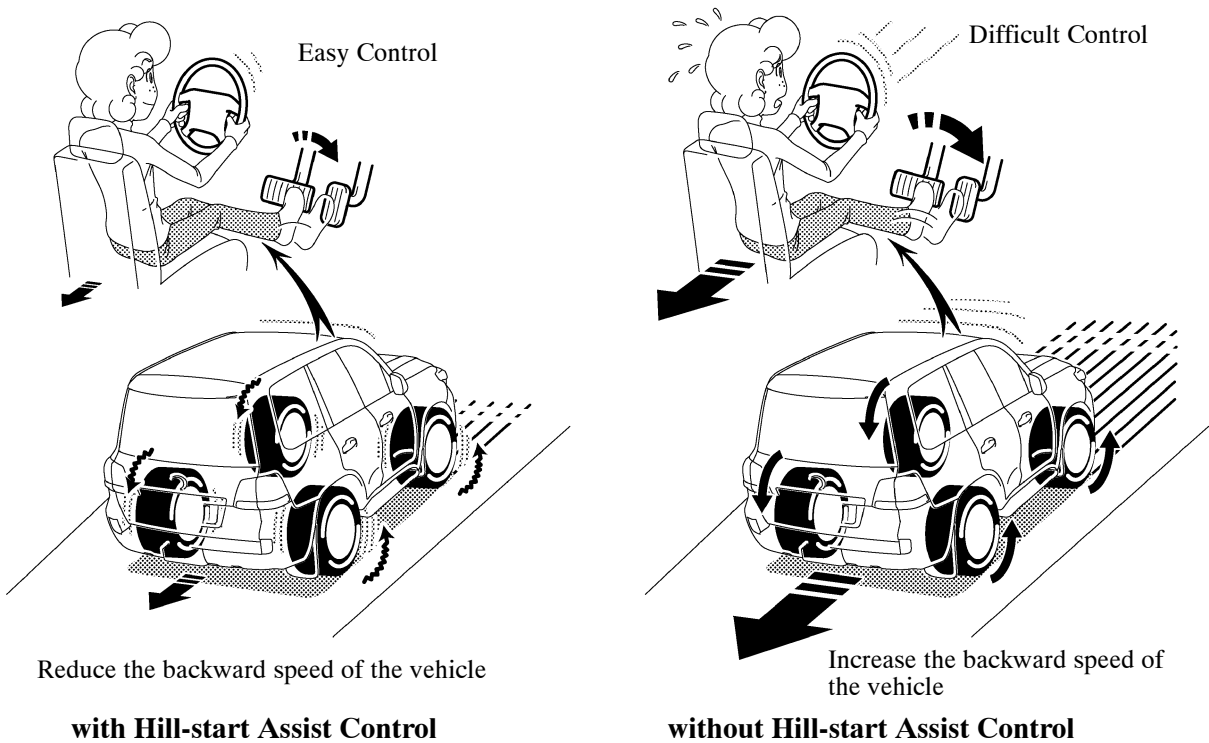


Making a Right Turn

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7. Outline of Hill-start Assist Control

- When the vehicle starts off on a steep or slippery hill, the vehicle could descend backward while the driver switches from the pedal brake to the accelerator pedal, thus making it difficult for the vehicle to start off. To prevent this from occurring, hill-start assist control temporarily (approximately 5 seconds at the maximum) applies the brakes to the 4-wheel in order to reduce the backward speed of the vehicle.
- Without hill-start assist control, the driver must quickly and precisely switch from the brake pedal to the accelerator pedal. With hill-start assist control however, the driver can start off easily and operate the pedal in a relaxed manner because hill-start assist control reduces the backward speed of the vehicle.



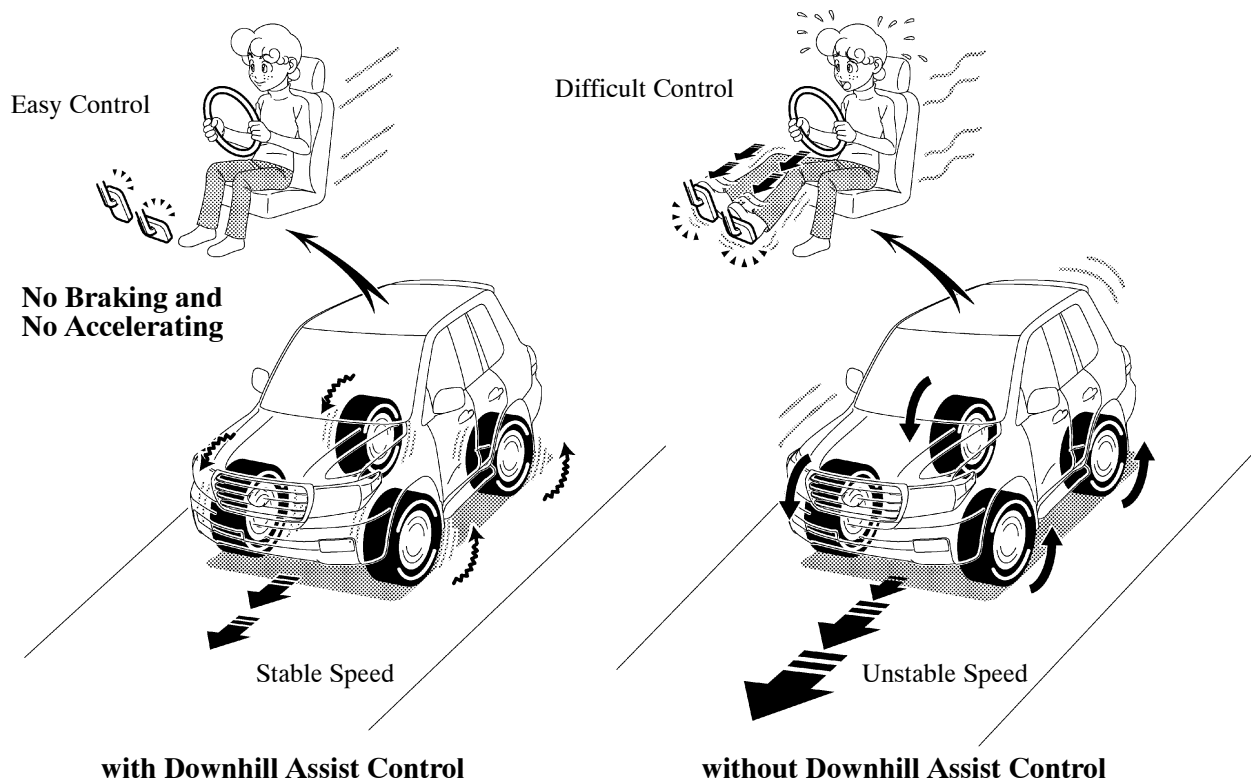
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- Hill-start assist control operates when all of the following conditions have been met:

Hill-start Assist Control Operate Condition	• Shift lever position is D or S positions. • The brake pedal is not pressed. • The skid control ECU has detected the backward movement of the vehicle when the driver is starting off on a hill.
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8. Outline of Downhill Assist Control (Only for 1VD-FTV engine models)

- When the vehicle is descending a steep hill and the engine brake alone cannot provide a sufficient deceleration force while the transfer is in the L4 range, downhill assist control effects 4-wheel brake control to maintain a constant, low vehicle speed. Thus, the vehicle is able to descend in a stable manner without causing the wheels to become locked.
- When the vehicle descends a steep hill without downhill assist control, the driver must pay close attention to the brake and accelerator pedal maneuvers. However, with downhill assist control, the driver can concentrate on the steering operation, without accelerator and brake pedals operation.
- Downhill assist control enables the vehicle to maintain stability because it can descend a slippery hill at low speeds without causing the wheels to become locked.



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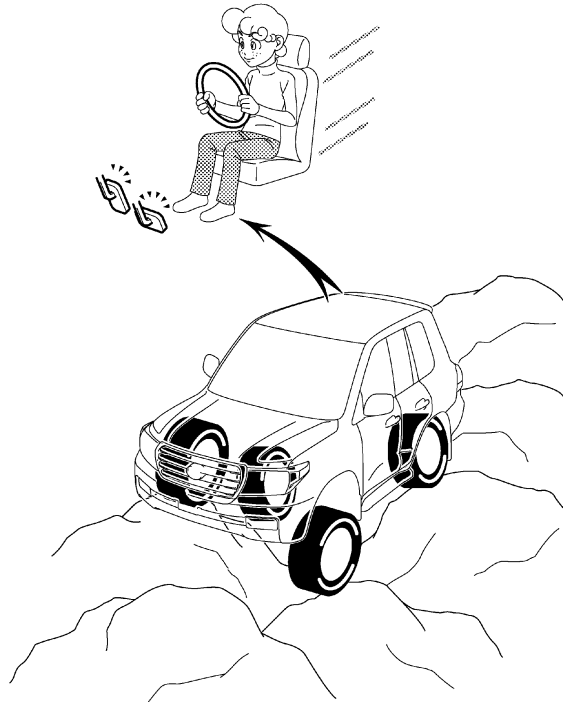
- Downhill assist control operates when all of the following conditions have been met:

Downhill Assist Control Operate Condition	• DAC switch ON • Transfer is L4 range. • Accelerator pedal and brake pedal are not pressed. • Descending a hill at a vehicle speed of 25 km/h (16 mph) or less.
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9. Outline of CRAWL (Only for 2UZ-FE engine models)

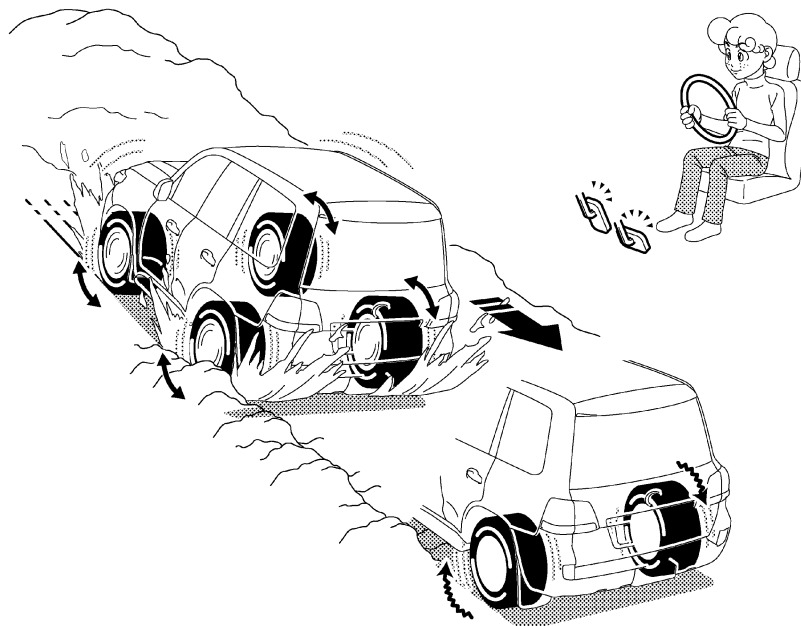
The CRAWL is a driving support system that regulates the engine output and the brake force, enabling the vehicle to be driven at a constant speed on rough ground or slippery roads.

- When the vehicle drives across rough grounds such as extremely bumpy rough roads, the driver needs to pay attention to the steering, braking and acceleration operations. However, with the CRAWL, the driver can concentrate on steering, without worrying about the accelerator or brake pedals.



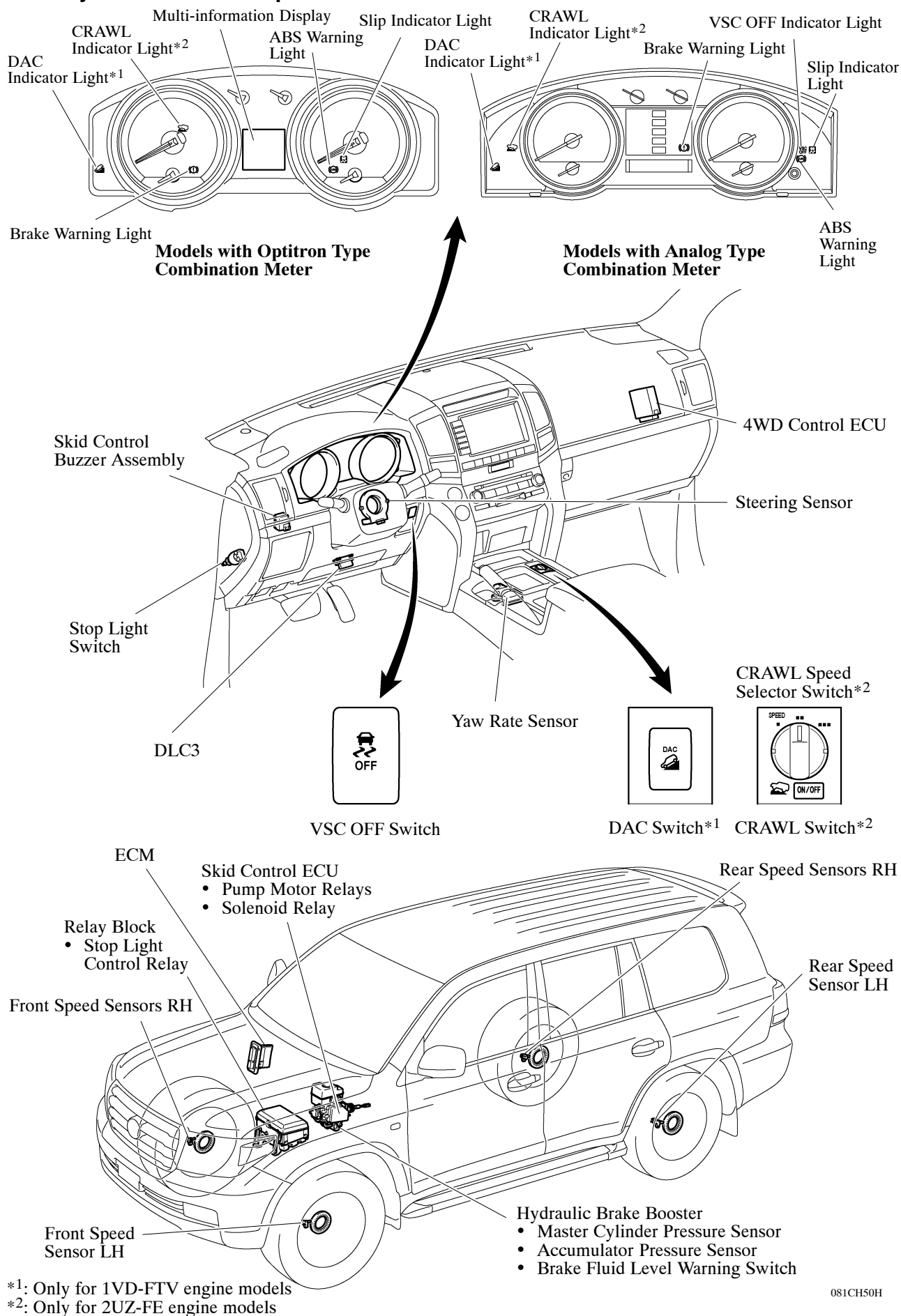
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- The CRAWL ensures good vehicle stability in situations such as when on a slippery or steep road because the chances of the wheels spinning or locking is minimized by regulating the engine output and the brake force.



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10. Layout of Main Components



11. Function of Main Components

Components		Function
Hydraulic Brake Booster (Including Brake Actuator)		- Assists with the brake pedal effort applied to the brake pedal. - Changes the brake fluid path based on the signals from the skid control ECU during the operations of the Multi-terrain ABS with EBD, brake assist, A-TRC, VSC, hill-start assist control, downhill assist control, and CRAWL in order to control the fluid pressure that is applied to the wheel cylinders. - For details about the parts, see page CH-169.
Combination Meter	Multi-information Display* ¹	Informs the driver of the status of the brake control system. For the items displayed on the multi-information display, see page CH-205.
	Brake Warning Light	- Lights up to alert the driver when a malfunction occurs in the EBD or skid control ECU. - Lights up to alert the driver that the hydraulic pressure of the accumulator in the hydraulic brake booster has decreased. - Lights up to alert the driver when the brake fluid in the reservoir tank reaches the low level
	ABS Warning Light	Lights up to alert the driver when the skid control ECU detects a malfunction in the Multi-terrain ABS, EBD or brake assist system.
	VSC OFF Indicator Light	Informs the driver of the standstill status of the VSC.
	Slip Indicator Light	Blinks to inform the driver when the A-TRC, VSC, hill-start assist control, downhill assist control, or CRAWL is operated.
	DAC Indicator Light* ²	Lights up to inform the driver when the DAC switch turned ON.
	CRAWL Indicator Light* ³	Lights up to inform the driver when the CRAWL switch turned ON.
Center Diff. Lock Indicator Light		Lights up to inform the driver when the center differential condition is locked.
Skid Control Buzzer Assembly		This buzzer has 3 types of sounds. - This buzzer sounds intermittently to inform the driver that the VSC and hill-start assist control are actives. - This buzzer sounds intermittently to inform the driver if the temperature of the brake actuator has increased excessively due to the continuous operation of the A-TRC, VSC, hill-start assist control, downhill assist control, and CRAWL. - This buzzer sounds continuously to alert the driver that the hydraulic pressure of the accumulator in the hydraulic brake booster has decreased, and that the power supply system is malfunctioning.
VSC OFF Switch		- By pressing the switch, only the A-TRC engine control is brought to a halt. - By pressing the switch for more than 3 seconds, both the A-TRC and VSC stop operating.
DAC Switch* ²		Turns the downhill assist control ON and OFF.
CRAWL Switch* ³		Turns the CRAWL ON and OFF.
CRAWL Speed Selector Switch* ³		Used to select the target vehicle speed when the CRAWL is active.

(Continued)

Components		Function
Skid Control ECU		Judges the vehicle driving condition based on signals from each sensor, and sends brake control signals to the brake actuator.
	Pump Motor Relays	Control the pump motor operation in the hydraulic brake booster.
	Solenoid Relay	Supply or cut off power to solenoid valves in the brake actuator.
Speed Sensor (4)		Detects the wheel speed and rotating direction of each of 4-wheel.
Steering Sensor		Detects the steering direction and angle of the steering wheel.
Yaw Rate Sensor		• Detects the vehicle's yaw rate. • Detects the vehicle's acceleration in the forward, rearward, and lateral.
Stop Light Control Relay		Turn on the stop light during hill-start assist control, downhill assist control or CRAWL operation.
Stop Light Switch		Detects the brake pedal depressing signal.
Park/Neutral Position Switch		Detects the shift position.
ECM	for 2UZ-FE Engine	Controls the throttle valve opening angle based on the signals receives from the skid control ECU, in order to control the engine output. Also, sends the throttle valve opening angle signal, accelerator pedal position signal, and engine speed signal to the skid control ECU.
	for 1VD-FTV Engine	Cutoff the fuel injection based on the signals receives from the skid control ECU, in order to control the engine output. Also, sends the accelerator pedal position signal, and engine speed signal to the skid control ECU.
Throttle Position Sensor (for 2UZ-FE engine)		Detects the opening of the throttle valve and inputs it into the ECM.
Throttle Control Motor (for 2UZ-FE engine)		Controls the opening of the throttle valve in accordance with the signals received from the ECM.
Accelerator Pedal Sensor (for 1VD-FTV engine)		Detects the depressing of the accelerator pedal and inputs it into the ECM.
EDU (for 1VD-FTV engine)		Controls the fuel injection in accordance with the signals received from the ECM.
Crankshaft Position Sensor		Detects the engine speed, and sends it via the ECM to the skid control ECU.
Transfer Shift Actuator	Center Diff. Lock Detection Switch	Detects the condition of the center differential lock.
4WD Control ECU		Detect the transfer condition shifted in the L4 and sends the L4 signal to the skid control ECU.

*1: Only on models equipped with the optitron type combination meter.

*2: Only for 1VD-FTV engine models.

*3: Only for 2UZ-FE engine models.

12. Construction of Main Components

Combination Meter

1) Warning and Indicator Lights and VSC Warning Buzzer

The warning and indicator lights and VSC warning buzzer operate as follows.

○: Illuminates (Continuous Sound) △: Blinks (Intermittent Sound) —: Not Available

Condition		Warning Light		Indicator Light				Buzzer
		BRAKE	ABS	VSC OFF*1	SLIP	DAC*2	CRAWL*3	
Initial Check		○	○	○	○	○	○	
During Operation	A-TRC				△			
	VSC				△			△
	Hill-start Assist Control				△			△
	Downhill Assist Control*2				△	○		
	CRAWL*3				△		○*4	
VSC OFF Switch	VSC OFF			○				
Driving Mode	Center Diff. Lock			○				
	Rear Diff. Lock		○	—	—	—	—	
During Actuator Protection					○	△	△	△
Cannot operate due to a malfunction	Multi-terrain ABS		○	○	○			
	EBD	○	○	○	○			
	Brake Assist		○	○	○			
	A-TRC			○	○			
	VSC			○	○			
	Hill-start Assist Control			○	○			
	Downhill Assist Control*2			○	○	△*5		
	CRAWL*3			○	○		△*6	
Others	Low Brake Fluid Level	○	○	○	○			
	ABS Malfunction		○	○	○	△*5	△*6	
	Low Accumulator Pressure	○	○	○	○			○
	Skid Control ECU Malfunction	○	○					

*1: Displayed on the multi-information display where equipped.

*2: Only for 1VD-FTV engine models.

*3: Only for 2UZ-FE engine models.

*4: When only the brake control of the CRAWL is operative, CRAWL indicator light blinks.

*5: Only when the DAC switch is pressed.

*6: If a malfunction occurs during the CRAWL operation, the CRAWL indicator light blinks and the CRAWL gradually stops operating. Then the CRAWL indicator light goes off. When the CRAWL is not operating, the indicator stays off.

2) Multi-information Display

The skid control ECU displays the status of the brake control system or system malfunctions on the multi-information display to inform the driver of the system status or malfunctions.

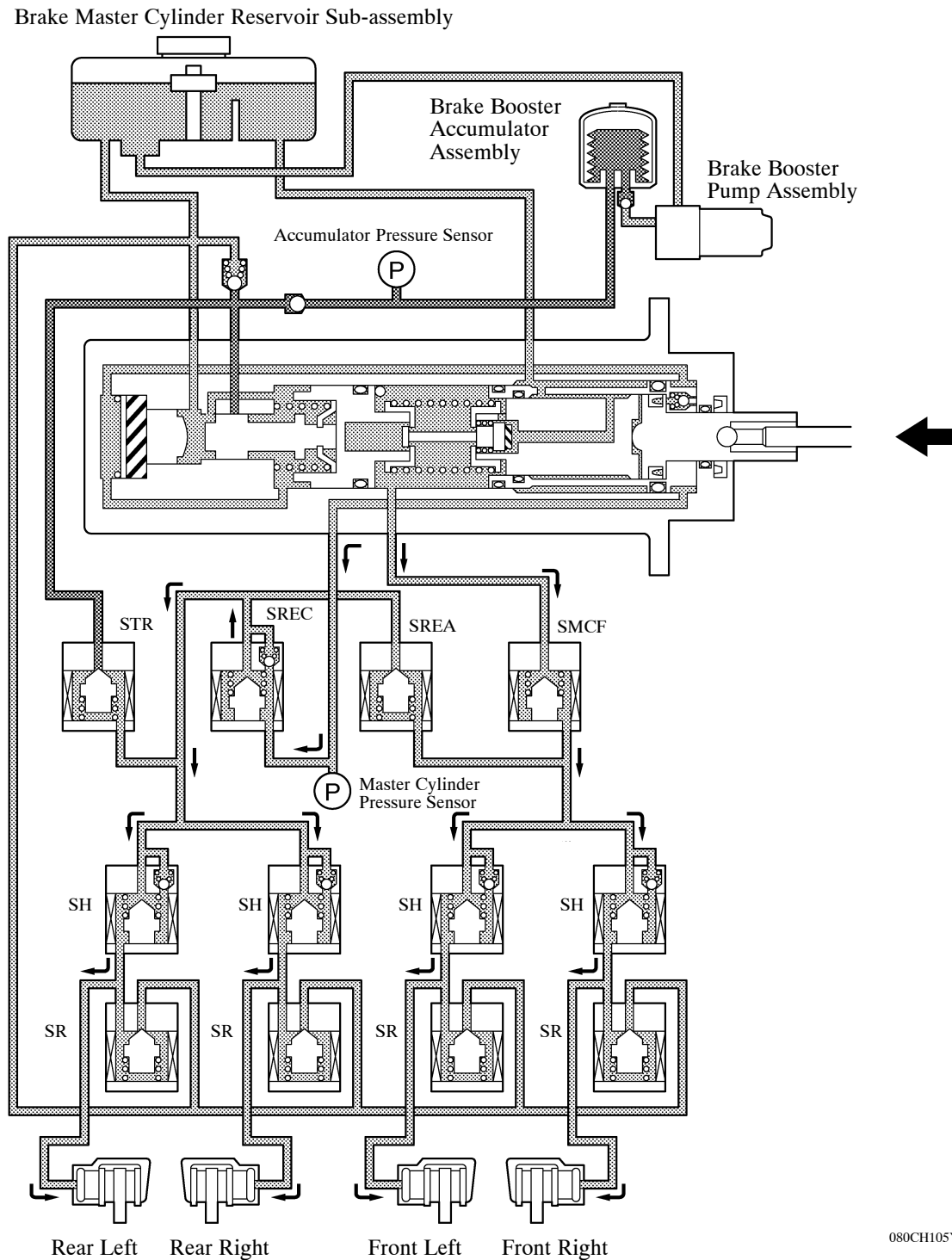
Display	Details
VSC OFF	Displayed when the A-TRC and VSC stop operating.
Check VSC	Displayed when a malfunction occurs in any one of the A-TRC, VSC, hill-start assist control, downhill assist control, or CRAWL. In addition, the master warning light and slip indicator light illuminate.
VSC Test Mode	Displayed when the VSC test mode is activated.
CRAWL is activated Speed: Lo Med Hi*	Displayed when the CRAWL switch is turned ON. Simultaneously, the target vehicle speed (Lo, Med or Hi), which is selected using the CRAWL speed selector switch, is displayed. In addition, the CRAWL indicator light illuminates and the slip indicator light blinks.
CRAWL has been deactivated*	Displayed when all the CRAWL controls stop operating. In addition, the CRAWL indicator light and the slip indicator light go off after blinking. If the CRAWL control is deactivated in any situation except when the CRAWL switch is turned OFF, the buzzer sounds for approximately 3 seconds.
	Displayed when only the CRAWL engine control stops operating. In addition, the slip indicator light goes off after blinking, and the buzzer sounds approximately 3 seconds. The CRAWL indicator light blinks.
CRAWL not available Select L4 and shift to D or R position*	Displayed when the CRAWL switch is turned ON in either of the following situations: - Transfer is in H4 range - Shift position is in P or N range
CRAWL not available Check System operation conditions*	Displayed when the CRAWL switch is turned ON in any one of the following situations: - Vehicle is running - Engine is not running - Parking brake is being applied - Brake pedal is not being depressed - Driver's door is open - 2nd start switch is turned ON - Brake actuator temperature is too high - ATF temperature is too high - CRAWL is malfunctioning

*: Only for 2UZ-FE engine models

13. System Operation

Normal Braking

During normal braking, all solenoid valves are turned OFF.



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SMCF: Master Cut for Front Solenoid Valve
 SREA: Regulator Apply for Front Solenoid Valve
 SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
 (Accumulator Cut Solenoid Valve)
 SH: Pressure Holding Solenoid Valve
 SR: Pressure Reduction Solenoid Valve

Multi-terrain ABS with EBD Operation

The detailed operation is the same as that of brake control system (Multi-terrain ABS with EBD). For details, see page CH-184.

Brake Assist Operation

The detailed operation is the same as that of brake control system (brake assist). For details, see page CH-188.

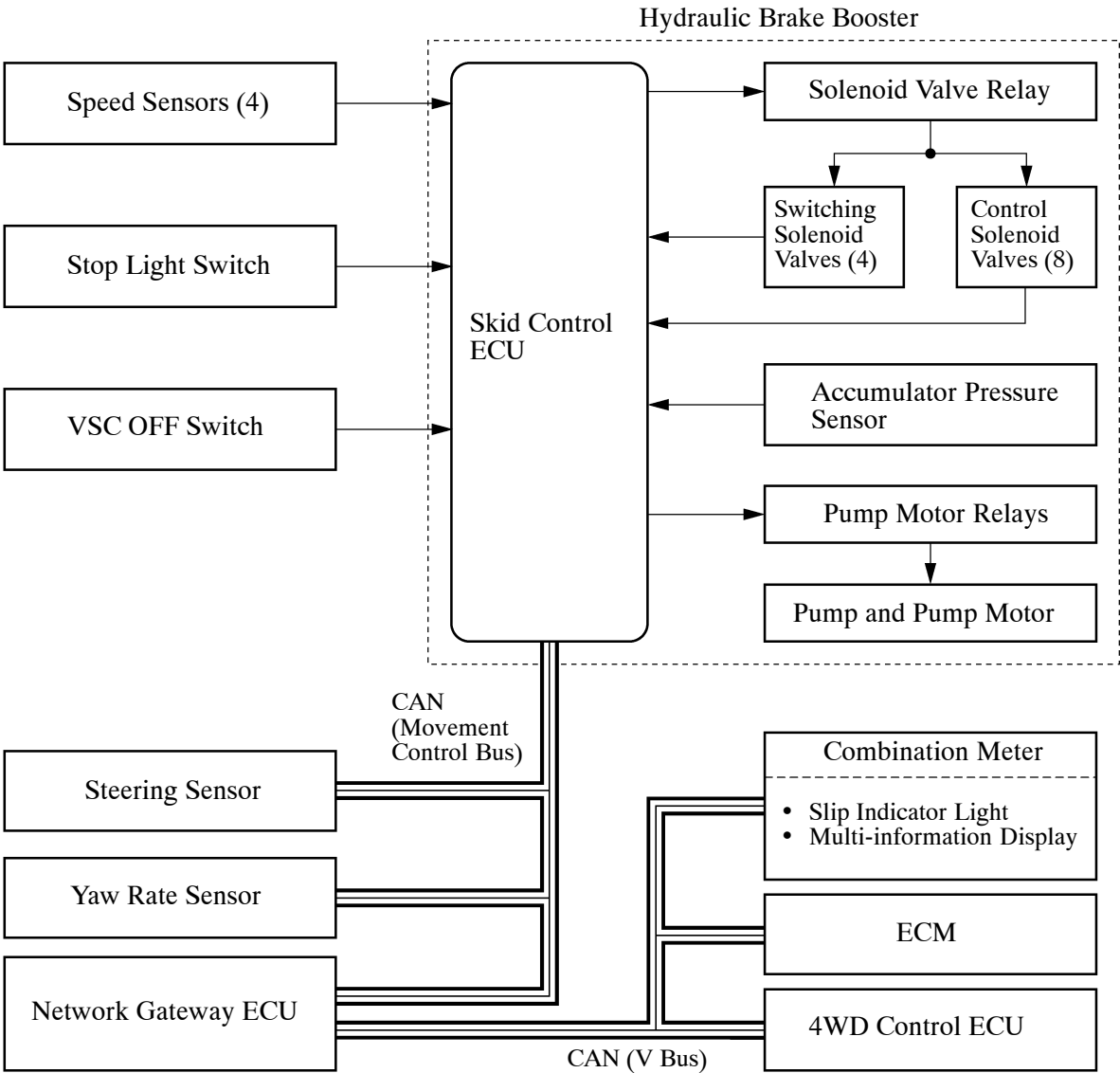
A-TRC Operation

1) General

- Based on the vehicle speed that has been calculated from each speed sensor and signals of the yaw rate sensor, the skid control ECU computes the target control speed in accordance with the transfer range.
- The skid control ECU outputs an A-TRC operate signal to the ECM and the combination meter. Upon receiving this signal, the ECM effects throttle control to regulate the engine output. On 1VD-FTV engine models, the engine output is regulated via fuel injection. The combination meter causes the slip indicator light to blink.

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► System Diagram ◀



- The engine output control of the A-TRC function varies in accordance with the range in which the transfer is engaged. When the transfer is engaged in the H4 range, this function effects engine output control that varies between stability-priority and drivability-priority in accordance with the amount of pedal effort applied to the accelerator pedal. When the transfer is engaged in the L4 range, it affects engine output control on a drivability-priority basis.
- The skid control ECU compares the target control speed and the speeds of the wheels to determine whether or not a slippage exists. Upon detecting a slippage, the skid control ECU controls the solenoid valve of the hydraulic brake booster to control the brake fluid pressure that is applied to the slipping wheel. When the wheel speed becomes lower than the target control speed, the skid control ECU stops controlling the brake fluid pressure.
- As shown in the table below, the target control speed and the brake fluid pressure control vary in accordance with the transfer range.

Road Condition	Transfer Range	Control	Contents	Outline
Ordinary Road	H4	Target Control Speed	Vehicle Speed + Slip Rate (H range set value)	Control designed to ensure the ease of driving on low-friction roads, dirt roads, and general roads.
		Brake Control	Gradual Fluid Pressure Control	
Rocky or Off-road	L4	Target Control Speed	Vehicle Speed + Slip Rate (L range set value)	Control designed for rugged off-road driving.
		Brake Control	Sudden fluid Pressure Control	
Downhill	L4 + 1st gear	Target Control Speed	Vehicle speed when deceleration slippage has been determined during downhill driving.	Designed for rugged, off-road downhill driving with the engine brake applied. It prevents the acceleration of the vehicle that could be caused by the release of the engine brake.
		Brake Control	Fluid pressure control to the front wheels	

NOTICE

Precaution on A-TRC Operation

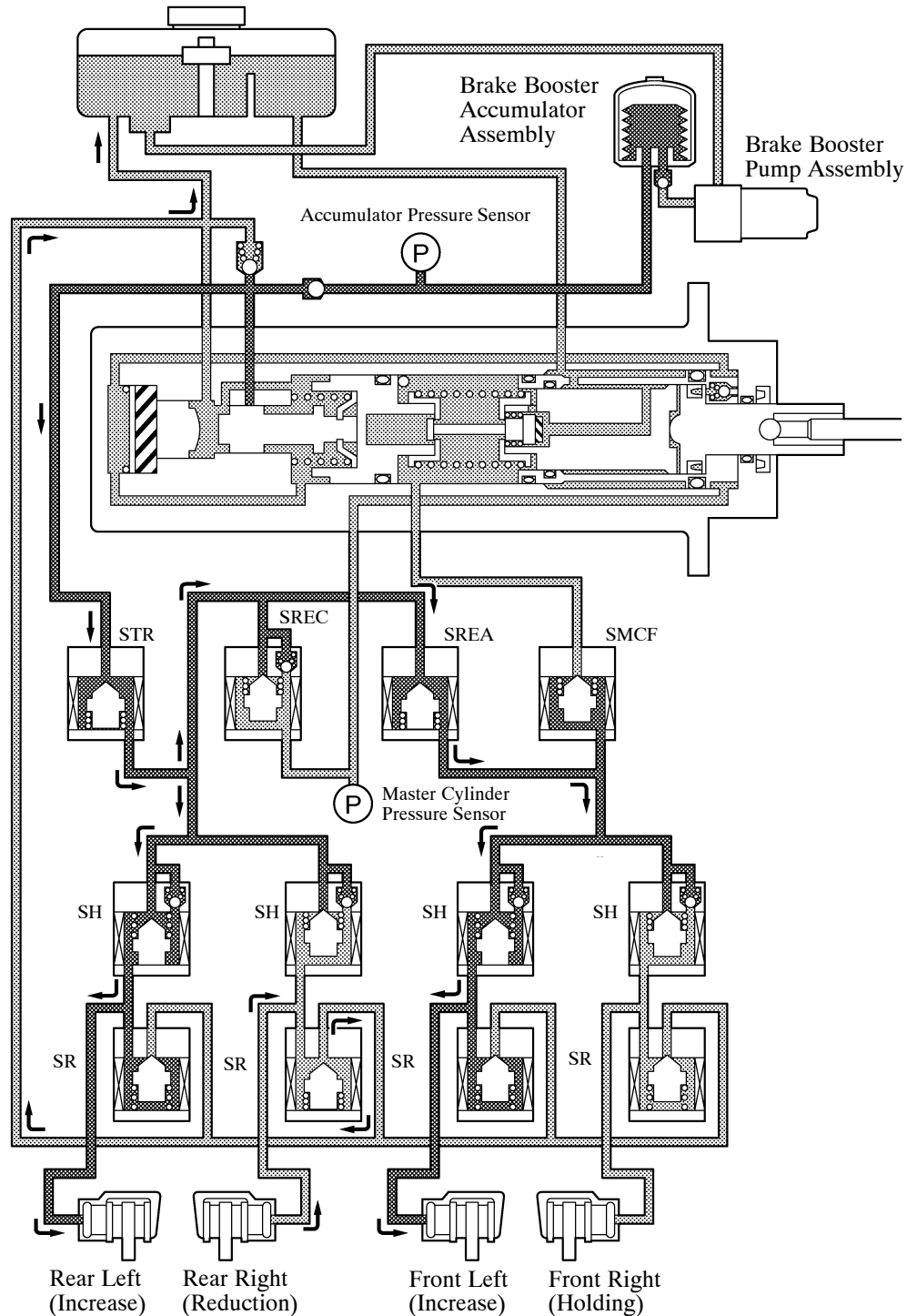
When the A-TRC is operating continuously while the vehicle is being driven on a slippery surface, the temperature of the actuator in the hydraulic brake booster increases. After a prescribed length of time elapses, the skid control ECU alerts the driver of this condition by causing the skid control buzzer to sound intermittently for approximately three seconds, and the slip indicator warning light to illuminate. Also, the A-TRC operation is momentarily interrupted to protect the actuator.

When the temperature of the actuator decreases, the slip indicator light turns OFF, and the A-TRC is automatically restored to an operating state.

2) A-TRC Operation

The fluid pressure control of the A-TRC independently controls the brake of each wheel by operating the individual solenoid valves in accordance with the signals received from the skid control ECU. The brake of each wheel is controlled in the following three modes: pressure reduction, pressure hold, and pressure increase modes.

Brake Master Cylinder Reservoir Sub-assembly



SMCF: Master Cut for Front Solenoid Valve
 SREA: Regulator Apply for Front Solenoid Valve
 SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
 (Accumulator Cut Solenoid Valve)
 SH: Pressure Holding Solenoid Valve
 SR: Pressure Reduction Solenoid Valve

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Item		A-TRC not Activated	A-TRC Activated		
			Mode		
			Increase	Holding	Reduction
Switching Solenoid Valve	SREC	OFF (Open)	ON (Close)	←	←
	STR	OFF (Close)	ON (Open)	←	←
	SMCF	OFF (Open)	ON (Close)	←	←
	SREA	OFF (Close)	ON (Open)	←	←
Control Switching Valve	Front Brake	SH	←	ON (Close)	←
		SR	←	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
	Rear Brake	SH	←	ON (Close)	←
		SR	←	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
				Reduce	Reduce

3) Cancel Operation

By pressing the VSC OFF switch, the A-TRC and VSC can be deactivated regardless of the status of the transfer gear and differential lock.

If the VSC or A-TRC is active when the vehicle attempts to escape from mud or snow, the engine output is limited despite the accelerator pedal being depressed, thus the escape will be difficult. The following switch operations enable the driver to deactivate or activate the VSC and A-TRC.

VSC OFF Switch Operation	Contents
Press the switch one time	The A-TRC engine control stops operating. The control can be resumed by pressing the switch again or when the vehicle speed increases.
Press and hold the switch for more than 3 seconds when the vehicle is stationary.	The A-TRC and VSC stop operating. These can be resumed by pressing the switch.

NOTE: When the transfer is in the L4 range and the center differential is not locked, even when both the A-TRC and VSC are not operating, vehicle speeds less than approximately 15km/h (9mph) may cause the A-TRC to be activated, however, this is not a malfunction.

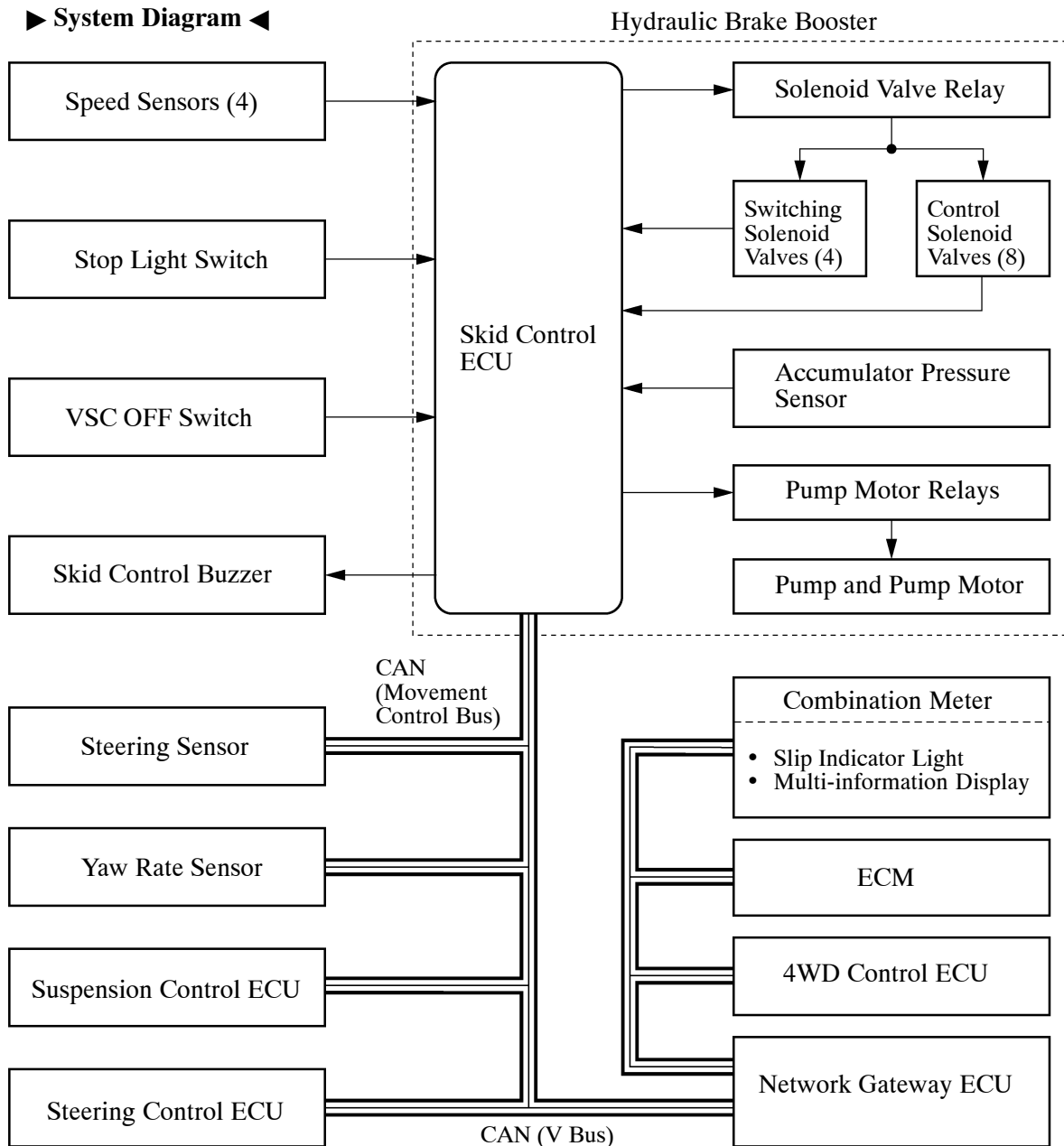
VSC Operation

1) General

- Based on the information provided by various sensors, switches, and the ECM, the skid control ECU determines the vehicle's yaw moment. Then, the skid control ECU controls the fluid pressure that is generated by the pump and pump motor and applies it by way of the solenoid valves to the brake wheel cylinder of each wheel in the following three modes: pressure reduction, pressure hold, and pressure increase modes. As a result, the tendency of the front wheels or the rear wheels to skid is restrained.
- At this time, the skid control ECU outputs a VSC operation signal to the ECM, the suspension control ECU and the combination meter cause the skid control buzzer to sound intermittently. Upon receiving the signal, the ECM regulates the engine output using fuel injection. The suspension control ECU regulates the suspension damping force. The combination meter causes the slip indicator light to blink.

NOTE: When the brake pedal is depressed while the VSC brake control is in operation, the brake pedal becomes more difficult to depress than it used to be, however, this is not a malfunction. By continuing to depress the brake pedal further, the appropriate brake force can be obtained.

► System Diagram ◀



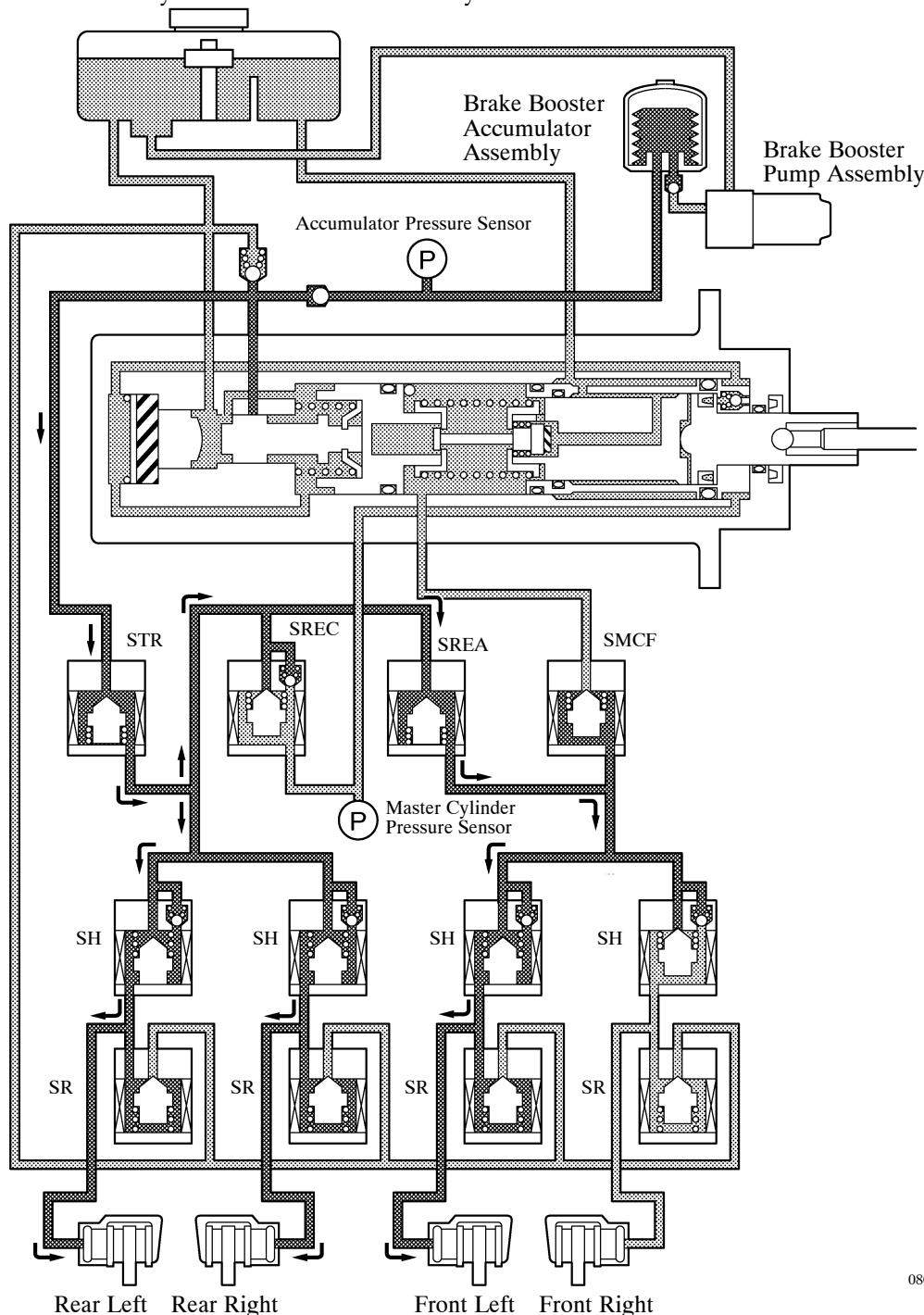
2) Front Wheel Skid Restraining (Turning to the Right)

In front wheel skid restraining control, the brakes of the rear wheels and front wheel of the outer side of the turn is applied. Also, depending on whether the brake is ON or OFF and the condition of the vehicle, there are circumstances in which the brake might not be applied to the wheels even if those wheels are targeted for braking. The diagram below shows the hydraulic circuit in the pressure increase mode, as it restrains a front wheel skid condition while the vehicle makes a right turn.

In other operating modes, the pressure holding valve and the pressure reduction valve are turned ON/OFF according to the Multi-terrain ABS with EBD operation pattern.

► VSC Operation (Front Wheel Skid Restraining) ◀

Brake Master Cylinder Reservoir Sub-assembly



080CH110Y

SMCF: Master Cut for Front Solenoid Valve
SREA: Regulator Apply for Front Solenoid Valve
SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
(Accumulator Cut Solenoid Valve)
SH: Pressure Holding Solenoid Valve
SR: Pressure Reduction Solenoid Valve

Item			VSC not Activated	VSC Activated			
				Mode			
				Increase	Holding	Reduction	
Switching Solenoid Valve	SREC		OFF (Open)	ON (Close)	←	←	
	STR		OFF (Close)	ON (Open)	←	←	
	Front Brake	SMCF		OFF (Open)	ON (Close)	←	←
		SREA		OFF (Close)	ON (Open)	←	←
SH		RH	OFF (Open)	ON (Close)	←	←	
		LH	OFF (Open)	←	ON (Close)	←	
SR		RH	OFF (Close)	←	←	←	
		LH	OFF (Close)	←	←	ON (Open)	
Wheel Cylinder Pressure		RH	—	—	—	—	
		LH	—	Increase	Hold	Reduce	
Control Switching Valve	Rear Brake	SH		OFF (Open)	←	ON (Close)	←
		SR		OFF (Close)	←	←	ON (Open)
		Wheel Cylinder Pressure		—	Increase	Hold	Reduce

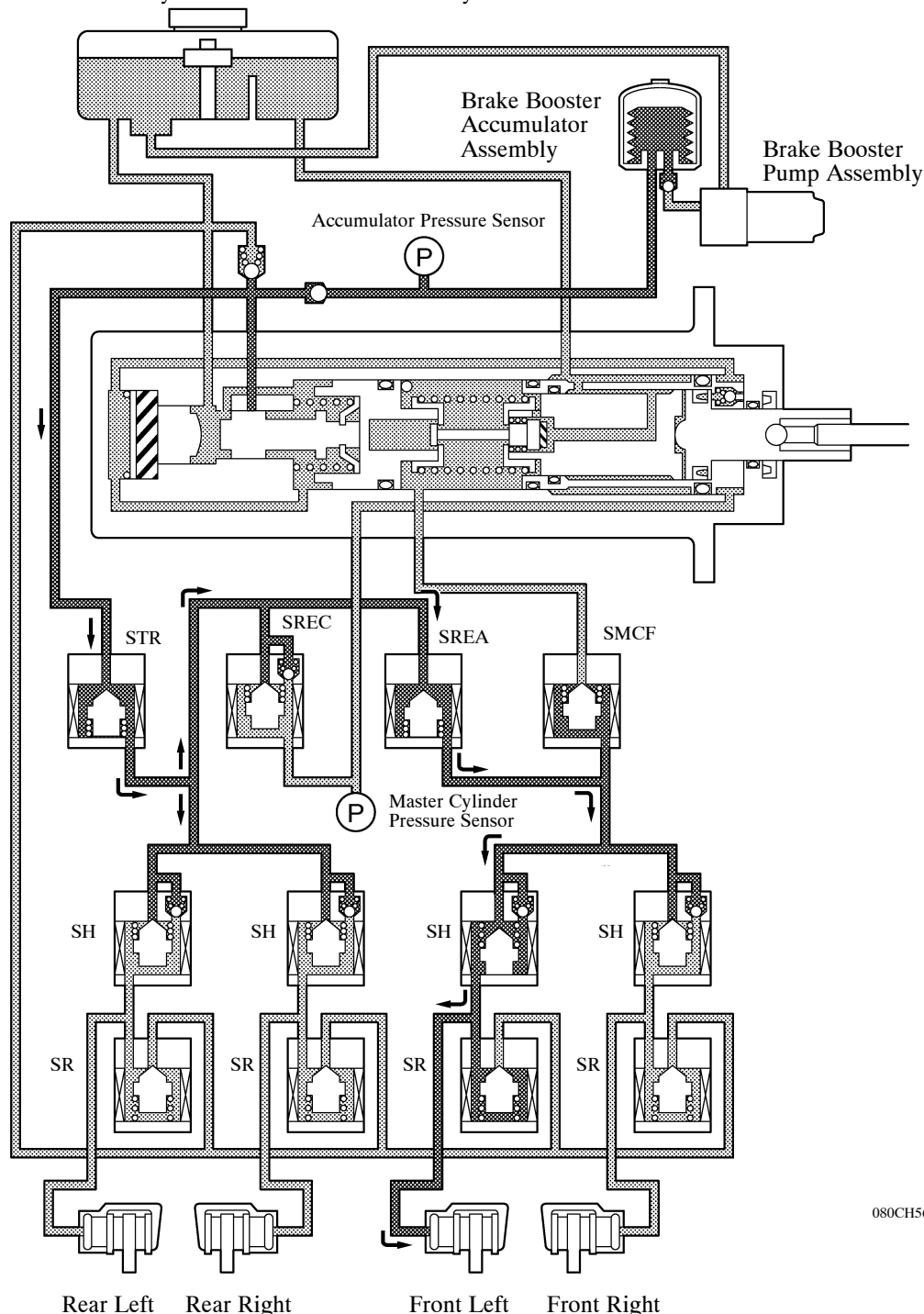
3) Rear Wheel Skid Restraining Control (Turning to the Right)

In rear wheel skid restraining control, the front wheel brake of the outer side of the turn is applied. Also, depending on whether the brake is ON or OFF and the condition of the vehicle, there are circumstances in which the brake might be applied to the rear brake of the outer side the turn. The diagram below shows the hydraulic circuit in the pressure increase mode, as it restrains a rear wheel skid condition while the vehicle makes a right turn.

In other operating modes, the pressure holding valve and the pressure reduction valve are turned ON/OFF according to the Multi-terrain ABS with EBD operation pattern.

► VSC Operation (Rear Wheel Skid Restraining) ◀

Brake Master Cylinder Reservoir Sub-assembly



080CH56Y

SMCF: Master Cut for Front Solenoid Valve
SREA: Regulator Apply for Front Solenoid Valve
SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
(Accumulator Cut Solenoid Valve)
SH: Pressure Holding Solenoid Valve
SR: Pressure Reduction Solenoid Valve

Item				VSC not Activated	VSC Activated			
					Mode			
					Increase	Holding	Reduction	
Switching Solenoid Valve	SREC			OFF (Open)	ON (Close)	←	←	
	STR			OFF (Close)	ON (Open)	←	←	
	Front Brake	SMCF			OFF (Open)	ON (Close)	←	←
		SREA			OFF (Close)	ON (Open)	←	←
SH		RH	OFF (Open)	ON (Close)	←	←		
		LH	OFF (Open)	←	ON (Close)	←		
SR		RH	OFF (Close)	←	←	←		
		LH	OFF (Close)	←	←	ON (Open)		
Control Switching Valve	Wheel Cylinder Pressure	RH	—	—	—	—		
		LH	—	Increase	Hold	Reduce		
	Rear Brake	SH			OFF (Open)	←	←	←
		SR			OFF (Close)	←	←	←
		Wheel Cylinder Pressure			—	—	—	—

4) Cancel Operation

By pressing the VSC OFF switch, the A-TRC and VSC can be deactivated regardless of the status of the transfer gear and differential lock.

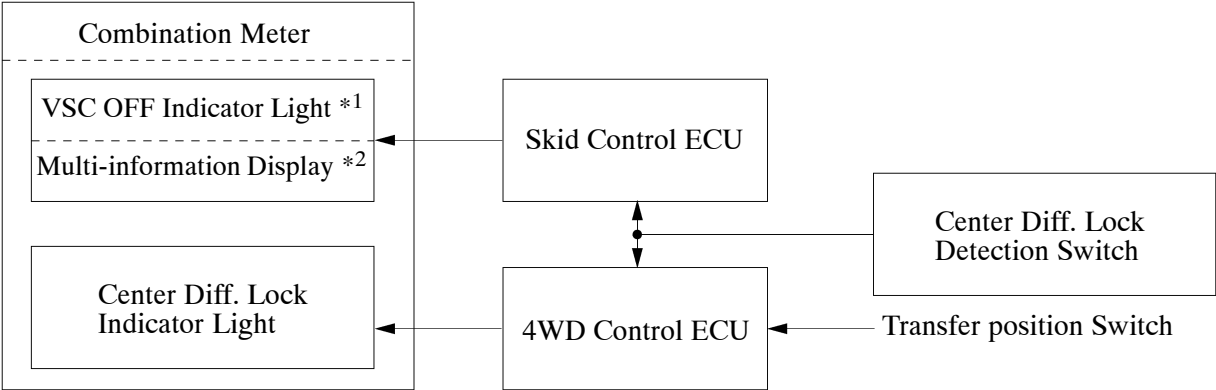
If the VSC or A-TRC is active when the vehicle attempts to escape from mud or snow, the engine output is limited despite the accelerator pedal being depressed, thus the escape will be difficult. The following switch operations enable the driver to deactivate or activate the VSC and A-TRC.

VSC OFF Switch Operation	Contents
Press the switch one time	The A-TRC engine control stops operating. The control can be resumed by pressing the switch again or when the vehicle speed increases.
Press and hold the switch for more than 3 seconds when the vehicle is stationary.	The A-TRC and VSC stop operating. These can be resumed by pressing the switch.

5) VSC Prohibit Control

When the center differential is locked, VSC is prohibited. At this time, the center differential indicator light and the VSC OFF indicator light* turn ON, or multi-information display shows VSC OFF. After the center differential is unlocked, the 2 indicator lights will turn OFF, and the VSC operation will turn ON.

► System Diagram ◀



*1: Only on models equipped with the analog type combination meter.
*2: Only on models equipped with the optitron type combination meter.

Hill-start Assist Control Operation

Based on the information provided by various sensors, switches, and the ECM, the skid control ECU computes the backward movement of the vehicle that occurs when the vehicle starts off on a hill. Then, the skid control ECU controls the fluid pressure that is generated by the pump and pump motor and applies it by way of the solenoid valves to the brake wheel cylinder of each wheel in the following three modes: pressure reduction, pressure hold, and pressure increase modes.

- The skid control ECU determines the state of the backward movement of the vehicle while the driver is attempting to drive uphill, based on the speed sensors and the park/ neutral position switch.
- This ECU determines the gradient of the hill, the acceleration state of the vehicle, the locked state and the direction of rotation of each wheel through the speed sensors, and the yaw rate sensor. Then, this ECU computes the amount of brake control that prevents the wheels from locking.
- During this operation, the skid control ECU outputs a hill-start assist control operation signal to the combination meter. This causes the slip indicator light to blink, outputs signals to the stop light control relay which turns ON the stop light.
- The hill-start assist control operates for approximately five seconds. At this time, skid control ECU informs the driver by the slow and intermittent sound of the skid control buzzer. After that, this ECU alerts the driver by using the quick and intermittent sound of the skid control buzzer, and gradually releases the brake hydraulic pressure in order to end the hill-start assist control operation.

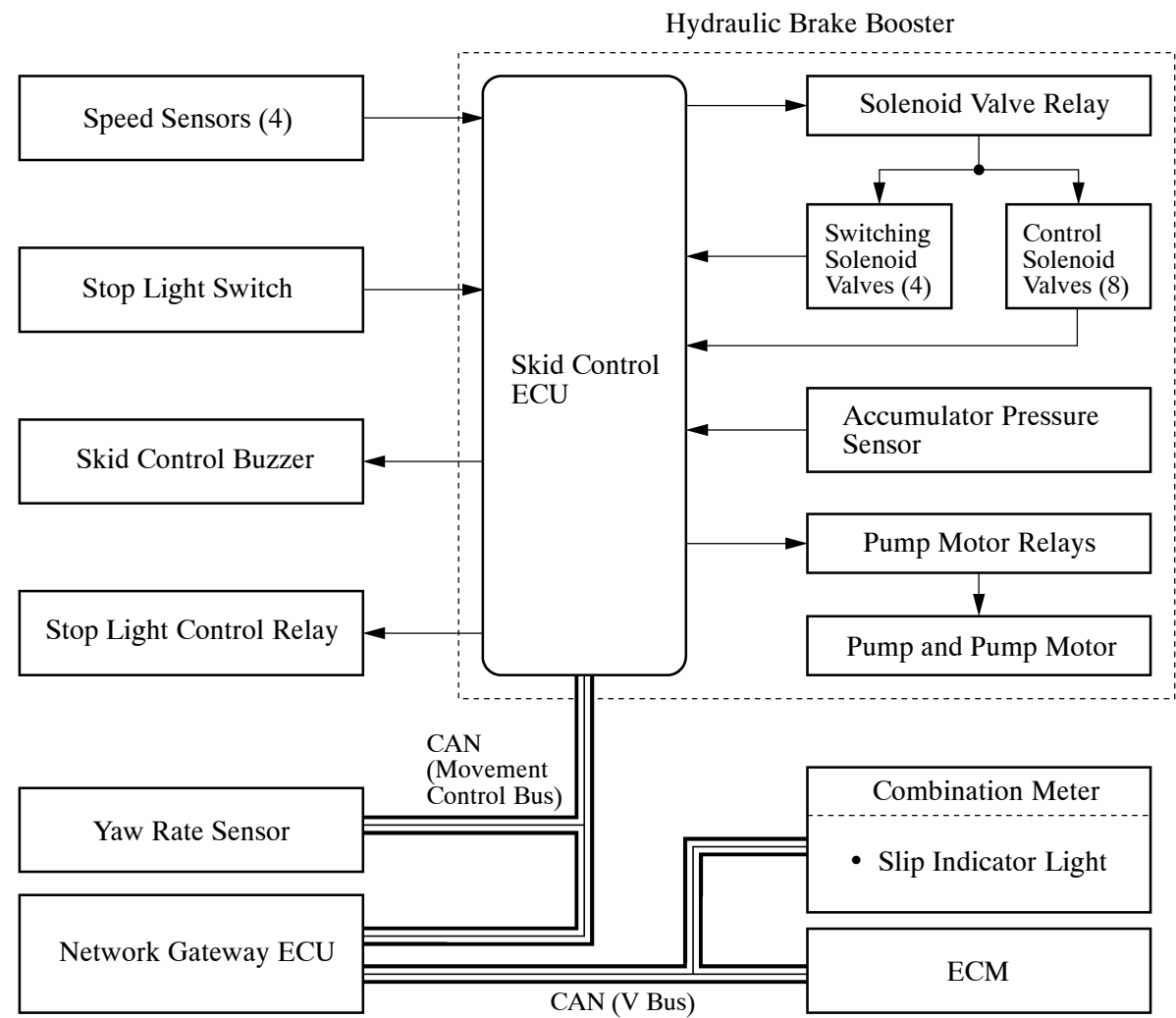
NOTICE

Precaution on Hill-start Assist Control Operation

When the hill-start assist control is operating continuously while the vehicle is being driven on a slippery surface, the temperature of the actuator in the hydraulic brake booster increases. After a prescribed length of time elapses, the skid control ECU alerts the driver of this condition by causing the slip indicator light to illuminate. Also, the hill-start assist control operation is momentarily interrupted to protect the actuator.

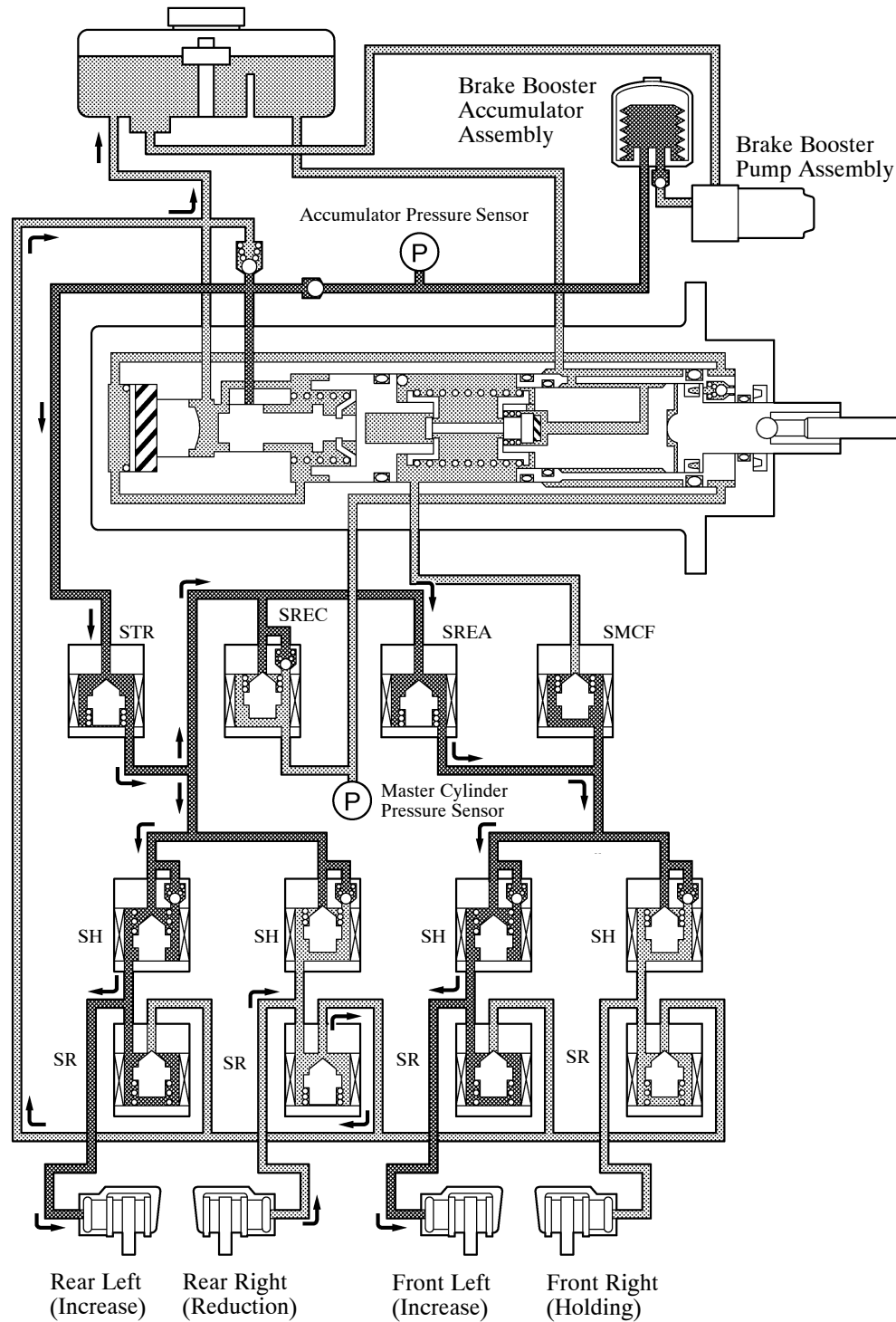
When the temperature of the actuator decreases, the slip indicator light turns OFF, and the hill-start assist control is automatically restored to an operating state.

► System Diagram ◀



► Hill-start Assist Control Operation ◀

Brake Master Cylinder Reservoir Sub-assembly



SMCF: Master Cut for Front Solenoid Valve
 SREA: Regulator Apply for Front Solenoid Valve
 SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
 (Accumulator Cut Solenoid Valve)
 SH: Pressure Holding Solenoid Valve
 SR: Pressure Reduction Solenoid Valve

080CH109Y

Item		Hill-start Assist Control not Activated	Hill-start Assist Control Activated		
			Mode		
			Increase	Holding	Reduction
Switching Solenoid Valve	SREC	OFF (Open)	ON (Close)	←	←
	STR	OFF (Close)	ON (Open)	←	←
	SMCF	OFF (Open)	ON (Close)	←	←
	SREA	OFF (Close)	ON (Open)	←	←
Control Switching Valve	Front Brake	SH	←	ON (Close)	←
		SR	←	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
	Rear Brake	SH	←	ON (Close)	←
		SR	←	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
				Reduce	Reduce

Downhill Assist Control Operation (Only for 1VD-FTV engine models)

Based on the information provided by various sensors, switches, and the ECM, the skid control ECU determines the conditions that enable downhill assist control operation. Then, the skid control ECU controls the fluid pressure that is generated by the pump and pump motor and applies it by way of the solenoid valves to the brake wheel cylinder of each wheel in the following three modes: pressure reduction, pressure hold, and pressure increase modes.

- The skid control ECU computes the vehicle speed, travel direction, and the gradient of the hill in accordance with the signals that are input by the speed sensor and the yaw rate sensor, and effects brake control to attain the target vehicle speed. The target vehicle speed is determined by the direction of the vehicle.

Travel Direction	Target Vehicle Speed
Forward	Approx. 4 to 9 km/h (2 to 6 mph)
Backward	Approx. 2 to 7 km/h (1 to 4 mph)

- During downhill assist control operation, the skid control ECU outputs signals to the stop light control relay to cause the stop light to turn ON and combination meter causing the slip indicator light to blink.
- The downhill assist control does not operate in the condition described below even if the DAC switch is turned ON; In this case, the DAC indicator light blinks to alert the driver.
 - If the DAC switch is turned OFF during a downhill assist control operation, the hydraulic pressure decreases gradually to end the downhill assist control operation.
 - Transfer is H4 range.
 - Driving at a vehicle speed of 25 km/h (16 mph) or higher.

NOTICE

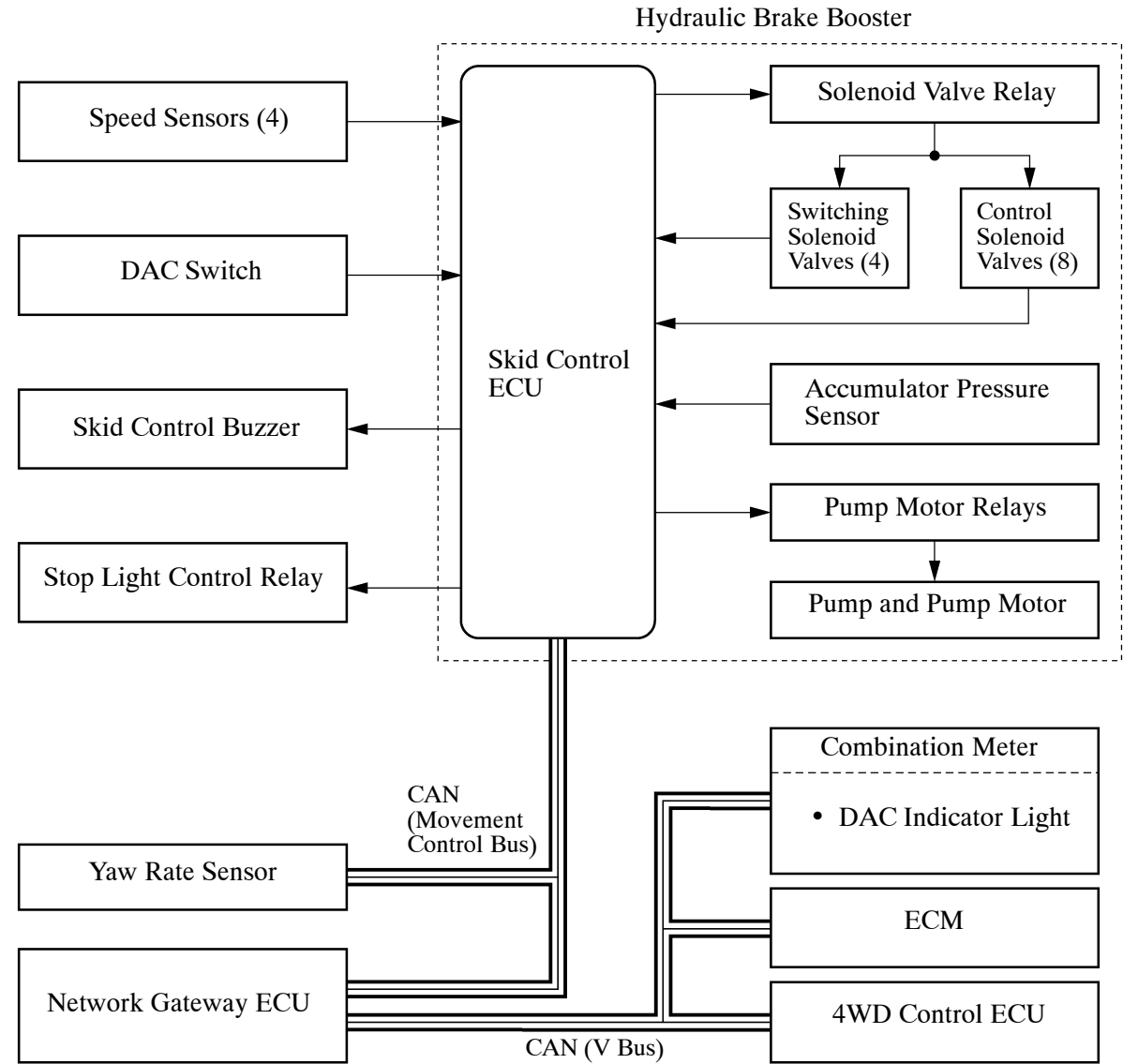
Precaution on Downhill Assist Control Operation

When driving on a steep downhill which causes the downhill assist control to be activated, select the L4 range and low gear, as well as use the engine brake. This enables the vehicle to run stably and reduce the load applied to the brake.

When the downhill assist control is operating continuously, the temperature of the actuator in the hydraulic brake booster increases. After a prescribed length of time elapses, the skid control ECU alerts the driver of this condition by causing the skid control buzzer to sound intermittently for approximately three seconds, slip indicator light to illuminate, and the DAC indicator light to blink. Also, the downhill assist control operation is momentarily interrupted to protect the actuator.

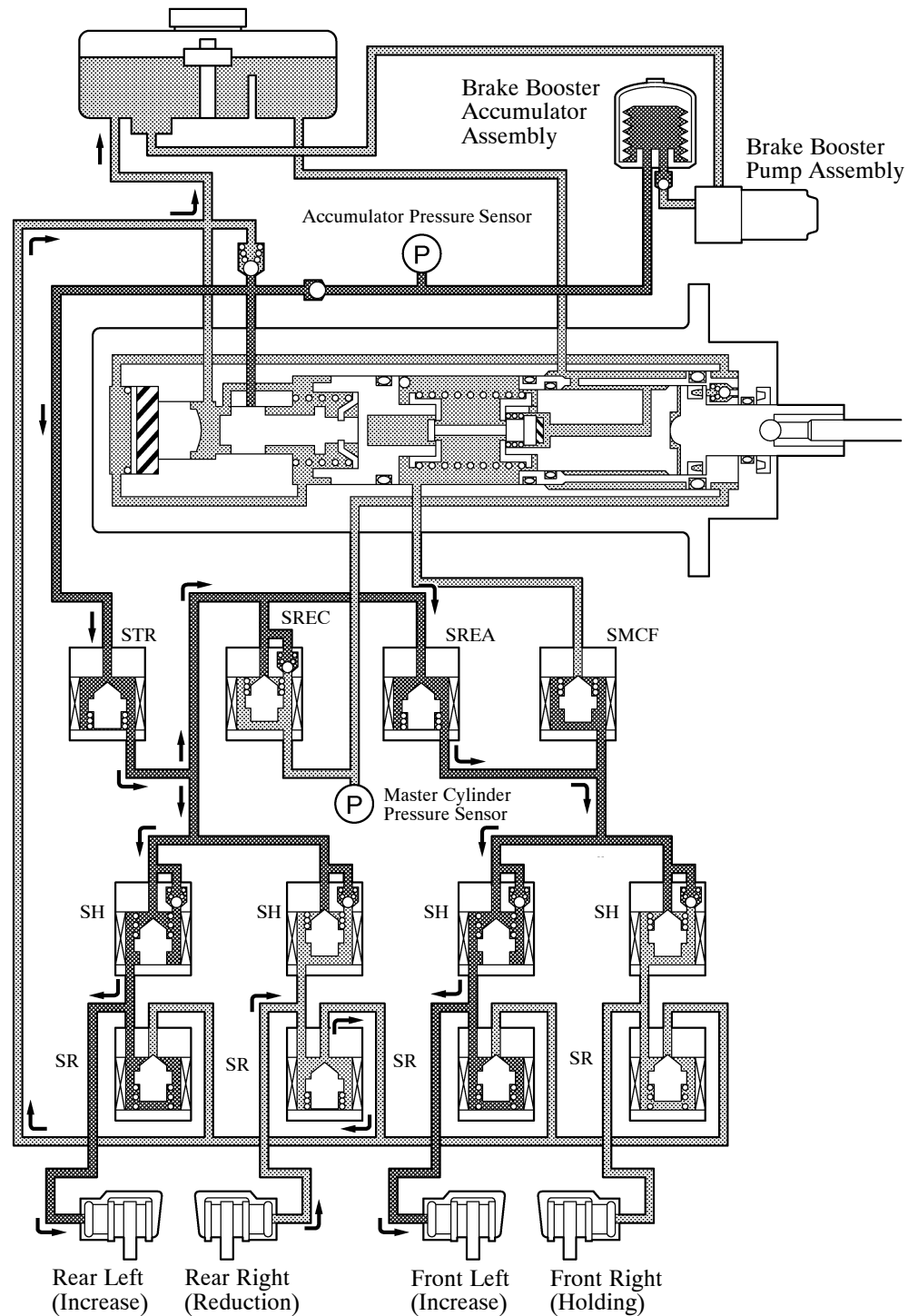
When the temperature of the actuator decreases, the slip indicator light turns OFF, and the downhill assist control is automatically restored to an operating state.

► System Diagram ◀



► Downhill Assist Control Operation ◀

Brake Master Cylinder Reservoir Sub-assembly



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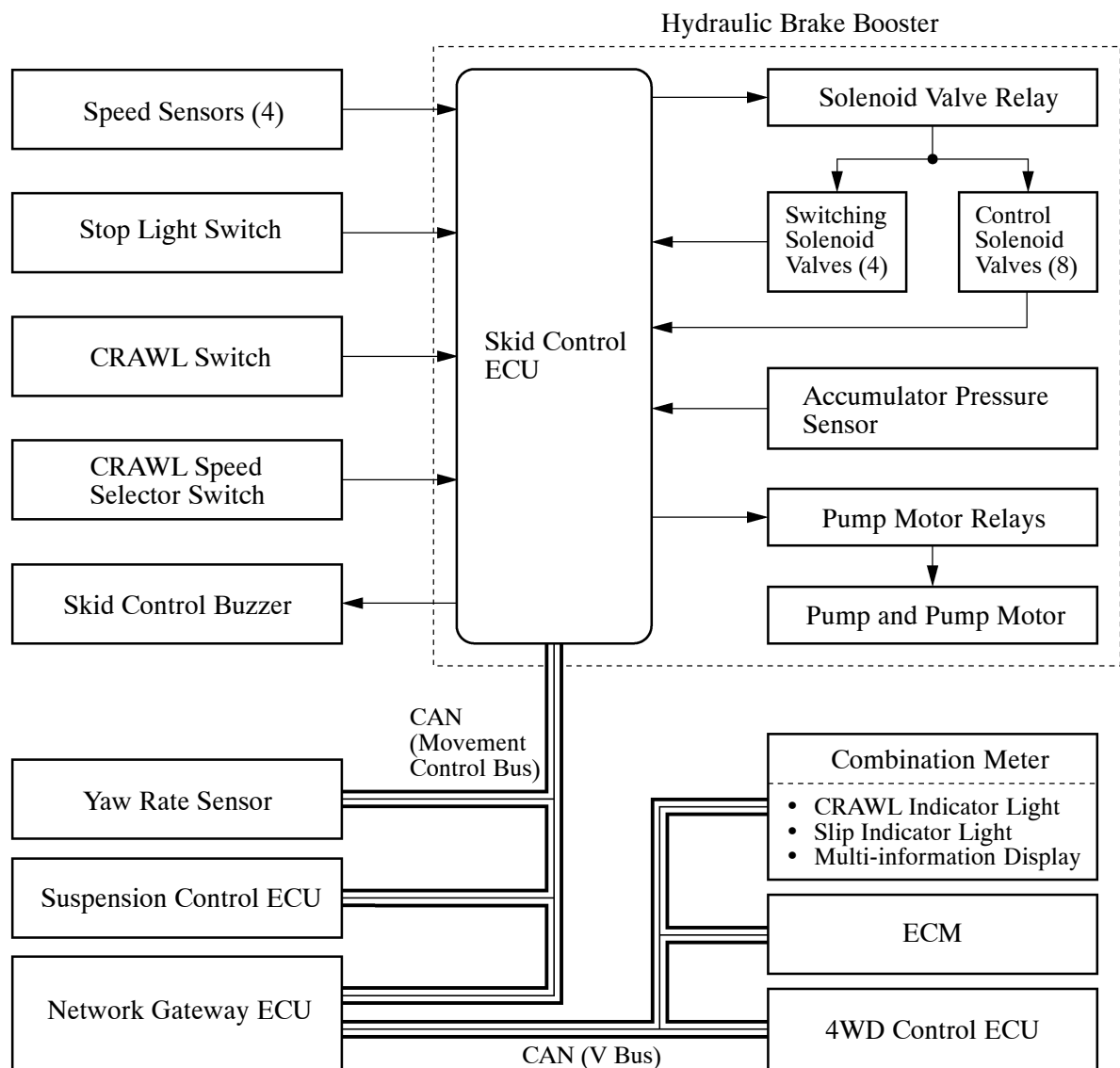
SMCF: Master Cut for Front Solenoid Valve
SREA: Regulator Apply for Front Solenoid Valve
SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
(Accumulator Cut Solenoid Valve)
SH: Pressure Holding Solenoid Valve
SR: Pressure Reduction Solenoid Valve

Item		Downhill Assist Control not Activated	Downhill Assist Control Activated		
			Mode		
			Increase	Holding	Reduction
Switching Solenoid Valve		SREC	OFF (Open)	ON (Close)	←
		STR	OFF (Close)	ON (Open)	←
		SMCF	OFF (Open)	ON (Close)	←
		SREA	OFF (Close)	ON (Open)	←
Control Switching Valve	Front Brake	SH	OFF (Open)	←	ON (Close)
		SR	OFF (Close)	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
	Rear Brake	SH	OFF (Open)	←	ON (Close)
		SR	OFF (Close)	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
					Reduce

CRAWL Operation (Only for 2UZ-FE engine models)**1) General**

- Based on the information provided by various sensors, switches, and the ECM, the skid control ECU determines the conditions that enable the CRAWL to operate. The skid control ECU computes the target vehicle speed (approximately 1 to 5km/h, 1 to 3mph) to control the vehicle speed, and the target wheel speeds to control the wheel slipping. The target vehicle speed can be changed using the speed selector switch.
- When the vehicle speed exceeds the target speed, the engine output is reduced. When the wheel speeds exceed the target speeds, the hydraulic brake pressure on the wheels is increased.
- When the vehicle speed falls below the target speed, the engine output is increased. When the wheel speeds fall below the target speeds, the hydraulic brake pressure on the wheels is reduced.

► System Diagram ◀**CH**

2) CRAWL Operation Condition

When the CRAWL switch is pressed with all of the following conditions met, the CRAWL control starts operating. The activation of the control causes the CRAWL indicator light to illuminate and the slip indicator light to blink.

- Transfer is L4.
- Shift position is in any position other than P or N.
- 2nd start switch is OFF.
- Vehicle speed is 0km/h (0mph).
- Brake pedal is being depressed.
- Parking brake is not applied.
- Driver's door is closed.
- Brake actuator temperature is too high.
- ATF temperature is too high.
- VSC system is operating normally.

3) CRAWL Operation Ending Condition

When any of the following conditions is met, the CRAWL control stops operating. The deactivation of the control causes the indicator light to go off. If the CRAWL control is deactivated in any situation except when the CRAWL switch is turned OFF, the buzzer sounds.

- CRAWL switch is turned OFF.
- Vehicle speed is above 25km/h (16mph).
- Transfer is changed to H4.
- Shift position is shifted into P.
- Brake actuator temperature is too high.
- VSC malfunctions. (If this happens, take appropriate actions such as stopping the vehicle.)

When any of the following conditions is met, only the CRAWL engine control is deactivated. Under this situation, only the CRAWL brake control continues to be active. When the CRAWL engine control is deactivated, the buzzer sounds intermittently and the CRAWL indicator light switches from illuminating to blinking. If this happens, take appropriate actions such as stopping the vehicle. When the CRAWL control stops operating completely, the indicator light goes off.

- Shift position is N.
- 2nd start switch is turned ON.
- Vehicle speed is above 10km/h (6mph).
- Brake pedal is depressed firmly.
- Parking brake is applied.
- ATF temperature is too high.
- Driver's door is opened.
- Engine stalls.
- Target vehicle speed cannot be reached and the skid control ECU determines that the CRAWL control cannot operate.

Prior to starting off, check the CRAWL indicator light. If the indicator light is blinking, the CRAWL brake control will work but the CRAWL engine control will not.

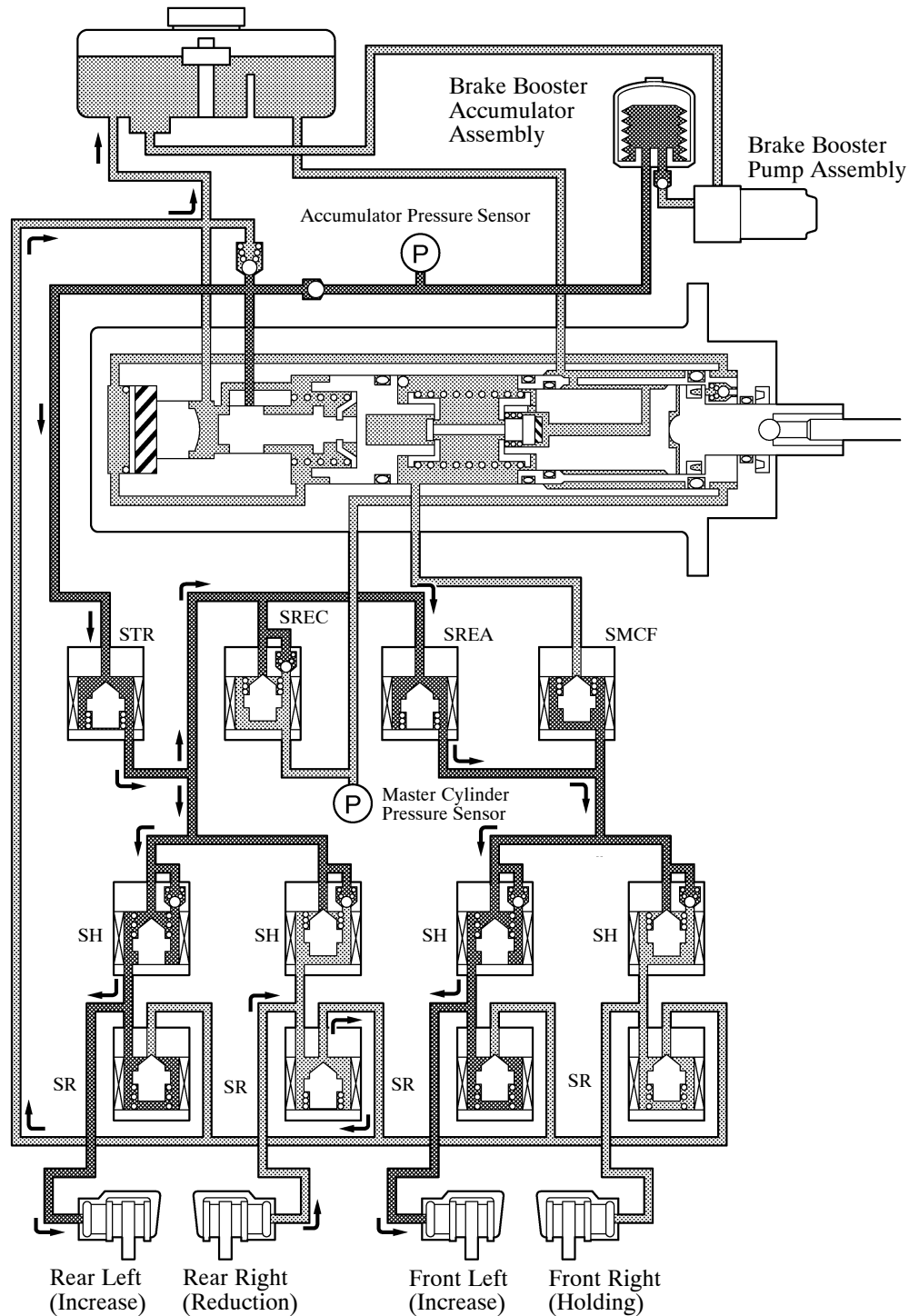
NOTICE**Precaution on CRAWL Operation**

- When the CRAWL is operating continuously, the temperature of the actuator in the hydraulic brake booster increases. After a prescribed length of time elapses, the skid control ECU alerts the driver of this condition by causing the skid control buzzer to sound intermittently for approximately three seconds, slip indicator light to illuminate, and the CRAWL indicator light to blink. Also, the CRAWL operation is gradually deactivated to protect the actuator.
When the temperature of the actuator decreases, the slip indicator light turns OFF, and the CRAWL becomes available.
- When the CRAWL continues operating, the ATF temperature may increase too much. If this happens, the ATF temperature warning light turns ON and the CRAWL operation stops operating to protect the automatic transmission. If the vehicle continues to be driven in this situation, ATF leakage or automatic transmission malfunctions may result. Therefore, immediately stop the vehicle in a safe area and select shift position P or N to let the ATF cool down enough.
When the temperature of ATF decreases, the ATF temperature warning light turns OFF and the CRAWL becomes available.

4) CRAWL Operation

The skid control ECU controls the fluid pressure that is generated by the pump and pump motor and applies it by way of the solenoid valves to the brake wheel cylinder of each wheel in the following three modes: pressure reduction, pressure hold, and pressure increase modes.

Brake Master Cylinder Reservoir Sub-assembly



080CH109Y

SMCF: Master Cut for Front Solenoid Valve
SREA: Regulator Apply for Front Solenoid Valve
SREC: Regulator Cut Solenoid Valve

STR: Traction Solenoid Valve
(Accumulator Cut Solenoid Valve)
SH: Pressure Holding Solenoid Valve
SR: Pressure Reduction Solenoid Valve

Item		CRAWL not Activated	CRAWL Activated		
			Mode		
			Increase	Holding	Reduction
Switching Solenoid Valve	Front Brake	SREC	OFF (Open)	ON (Close)	←
		STR	OFF (Close)	ON (Open)	←
		SMCF	OFF (Open)	ON (Close)	←
		SREA	OFF (Close)	ON (Open)	←
Control Switching Valve	Front Brake	SH	OFF (Open)	←	ON (Close)
		SR	OFF (Close)	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold
	Rear Brake	SH	OFF (Open)	←	ON (Close)
		SR	OFF (Close)	←	ON (Open)
		Wheel Cylinder Pressure	—	Increase	Hold

Engine Output Control

During a A-TRC, VSC, or CRAWL operation, the skid control ECU outputs a brake control operation signal to the ECM. Upon receiving this signal, the ECM effects throttle control to regulate the engine output. On 1VD-FTV engine models, the engine output is regulated via fuel injection.

Initial Check

Each time the engine switch is turned IG-ON, and the vehicle reaches approximately speed of 6 km/h (4 mph) or more, the skid control ECU performs the initial check.

The functions of each solenoid valve in the hydraulic brake booster are checked in order.

14. Diagnosis

- If the skid control ECU detects a malfunction in the brake control system (Multi-terrain ABS with EBD, brake assist, A-TRC, VSC, hill-start assist control, downhill assist control, CRAWL), then the ABS warning light, brake warning light, VSC OFF indicator light*, or slip indicator light illuminate either individually or as a group according to the function in which the malfunction has been detected, to alert the driver of the malfunction.
- At the same time, DTC (Diagnostic Trouble Code) is stored in memory of the skid control ECU. This DTC can be read by the following two methods.
 - The 5-digit DTC can be read by connecting an intelligent tester II to the DLC3.
 - The 2-digit DTC can be read from number of the blinking of ABS warning light by connecting the SST (90843-18040) to the TC and CG terminals of the DLC3.
- This system has a sensor signal check (test mode) function. This function is activated by connecting an intelligent tester II or by connecting the SST (09843-18040) to the TS and CG terminals of the DLC3. If the skid control ECU detects a malfunction during a sensor signal check (test mode), it stores the DTC in its memory.

For details on the DTC that are stored in skid control ECU memory and the DTC that are output through the sensor signal check function, see the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

*: Displayed on the multi-information display where equipped.

15. Fail-safe

If a failure occurs in the skid control ECU, sensors, or brake actuators, the system continues effecting brake control by excluding the failed area and using only the areas that are operating normally.